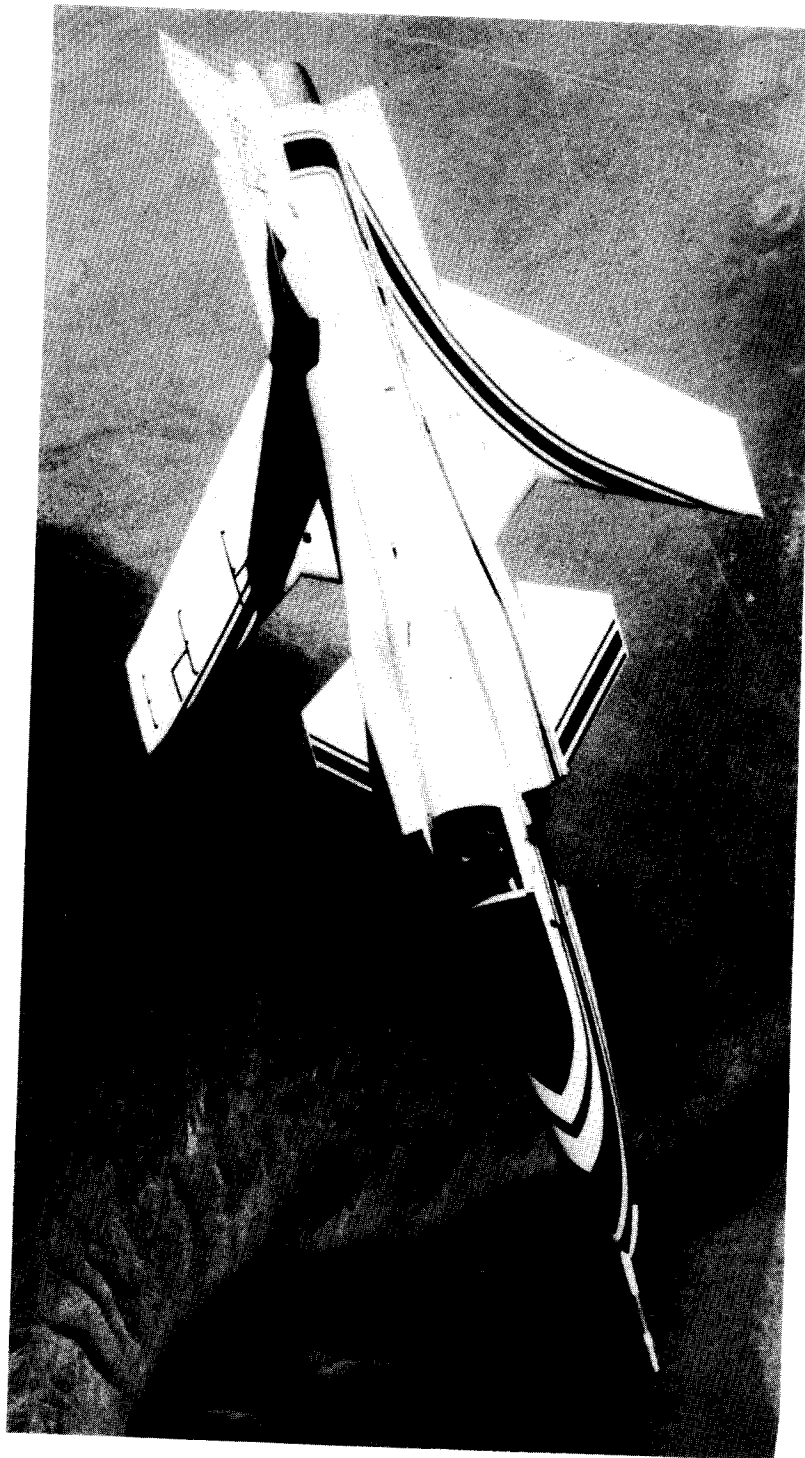




X-29 Forward-Swept-Wing Demonstrator

AIRFOIL AND GEOMETRY SELECTION

4.1 INTRODUCTION

Before the design layout can be started, values for a number of parameters must be chosen. These include the airfoil(s), the wing and tail geometries, wing loading, thrust-to-weight or horsepower-to-weight ratio, estimated takeoff gross weight and fuel weight, estimated wing, tail, and engine sizes, and the required fuselage size. These are discussed in the next three chapters.

This chapter covers selecting the airfoil and the wing and tail geometry. Chapter 5 addresses estimation of the required wing loading and thrust-to-weight ratio (horsepower-to-weight ratio for a propeller aircraft). Chapter 6 provides a more refined method for initial sizing than the quick method presented in the last chapter, and concludes with the use of the sizing results to calculate the required wing and tail area, engine size, and fuselage volume.

4.2 AIRFOIL SELECTION

The airfoil, in many respects, is the heart of the airplane. The airfoil affects the cruise speed, takeoff and landing distances, stall speed, handling qualities (especially near the stall), and overall aerodynamic efficiency during all phases of flight.

Much of the Wright Brothers' success can be traced to their development of airfoils using a wind tunnel of their own design, and the in-flight validation of those airfoils in their glider experiments of 1901-1902. The P-51 was regarded as the finest fighter of World War II in part because of its radical laminar-flow airfoil. Recently, the low-speed airfoils developed by Peter Lissaman contributed to the success of the man-powered Gossamer Condor.

Airfoil Geometry

Figure 4.1 illustrates the key geometric parameters of an airfoil. The front of the airfoil is defined by a leading-edge radius which is tangent to the upper and lower surfaces. An airfoil designed to operate in supersonic flow will have a sharp or nearly-sharp leading edge to prevent a drag-producing bow shock. (As discussed later, wing sweep may be used instead of a sharp leading edge to reduce the supersonic drag.)

The chord of the airfoil is the straight line from the leading edge to the trailing edge. It is very difficult to build a perfectly sharp trailing edge, so most airfoils have a blunt trailing edge with some small finite thickness.

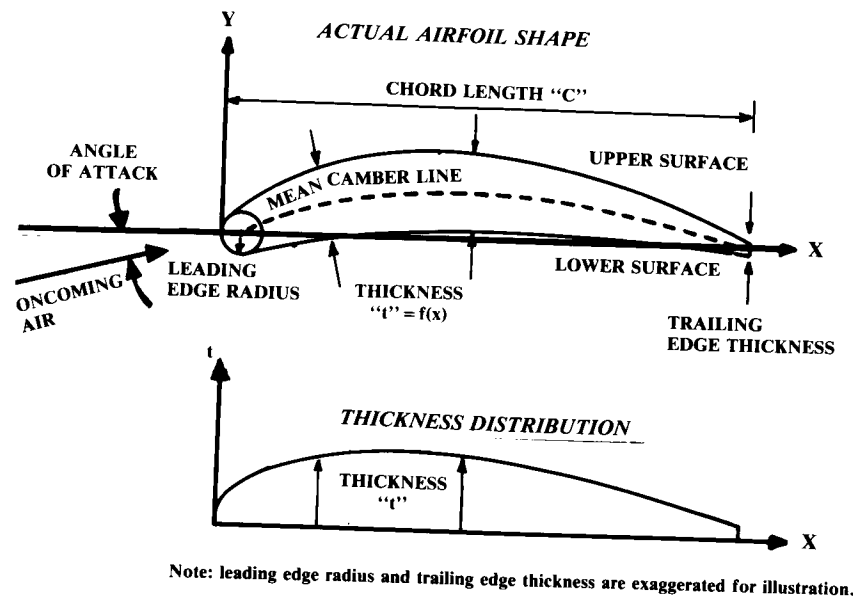


Fig. 4.1 Airfoil geometry.

"Camber" refers to the curvature characteristic of most airfoils. The "mean camber line" is the line equidistant from the upper and lower surfaces. Total airfoil camber is defined as the maximum distance of the mean camber line from the chord line, expressed as a percent of the chord.

In earlier days, most airfoils had flat bottoms, and it was common to refer to the upper surface shape as the "camber." Later, as airfoils with curved bottoms came into usage, they were known as "double-cambered" airfoils. Also, an airfoil with a concave lower surface was known as an "under-cambered" airfoil. These terms are technically obsolete but are still in common usage.

The thickness distribution of the airfoil is the distance from the upper surface to the lower surface, measured perpendicular to the mean camber line, and is a function of the distance from the leading edge. The "airfoil thickness ratio" (t/c) refers to the maximum thickness of the airfoil divided by its chord.

For many aerodynamic calculations, it has been traditional to separate the airfoil into its thickness distribution and a zero-thickness camber line. The former provides the major influence on the profile drag, while the latter provides the major influence upon the lift and the drag due to lift.

When an airfoil is scaled in thickness, the camber line must remain unchanged, so the scaled thickness distribution is added to the original camber line to produce the new, scaled airfoil. In a similar fashion, an airfoil which is to have its camber changed is broken into its camber line and thickness distribution. The camber line is scaled to produce the desired maximum camber; then the original thickness distribution is added to obtain the new

airfoil. In this fashion, the airfoil can be reshaped to change either the profile drag or lift characteristics, without greatly affecting the other.

Airfoil Lift and Drag

An airfoil generates lift by changing the velocity of the air passing over and under itself. The airfoil angle of attack and/or camber causes the air over the top of the wing to travel faster than the air beneath the wing.

Bernoulli's equation shows that higher velocities produce lower pressures, so the upper surface of the airfoil tends to be pulled upward by lower-than-ambient pressures while the lower surface of the airfoil tends to be pushed upward by higher-than-ambient pressures. The integrated differences in pressure between the top and bottom of the airfoil generate the net lifting force.

Figure 4.2 shows typical pressure distributions for the upper and lower surfaces of a lifting airfoil at subsonic speeds. Note that the upper surface of the wing contributes about two-thirds of the total lift.

Figure 4.3a illustrates the flowfield around a typical airfoil as a number of airflow velocity vectors, with the vector length representing local velocity magnitude. In Fig. 4.3b, the freestream velocity vector is subtracted from each local velocity vector, leaving only the change in velocity vector caused by the presence of the airfoil. It can be seen that the effect of the airfoil is to introduce a change in airflow, which seems to circulate around the airfoil in a clockwise fashion if the airfoil nose is to the left.

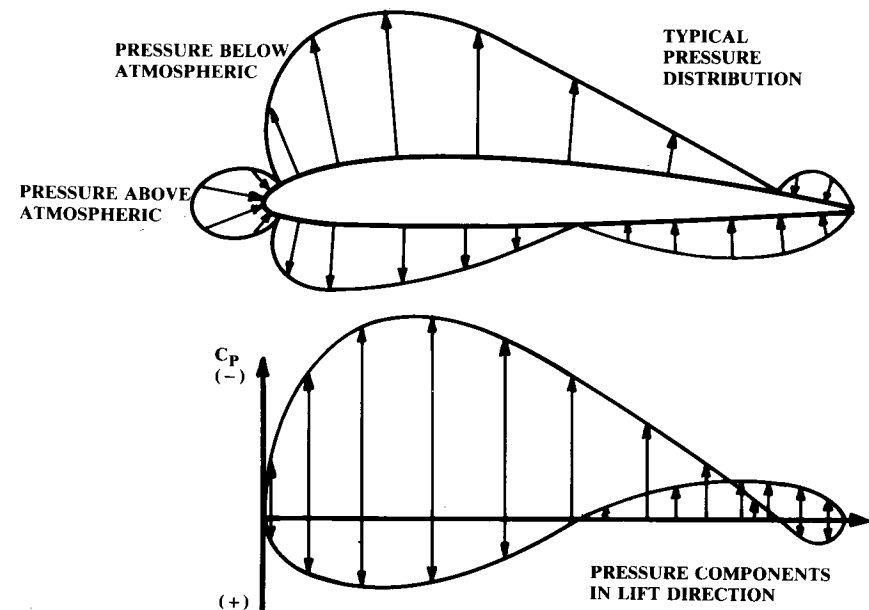


Fig. 4.2 Typical airfoil pressure distribution.

This "circulation" is the theoretical basis for the classical calculation of lift and drag-due-to-lift. The greater the circulation, the greater the lift. Circulation is usually represented by Γ and is shown as a circular flow direction as in Fig. 4.3c.

As a side effect of the generation of lift, the airfoil imparts a downward momentum on the flowfield. The downward force exerted on the air must equal the lift force produced on the airfoil. However, the lift is not "caused" by the downward motion of the air but by the pressure forces the air exerts upon the airfoil.

A flat board at an angle to the oncoming air will produce lift. However, the air going over the top of the flat "airfoil" will tend to separate from the surface, thus disturbing the flow and therefore reducing lift and greatly increasing drag (Fig. 4.4). Curving the airfoil (i.e., camber) allows the air flow to remain attached, thus increasing lift and reducing drag. The camber also increases lift by increasing the circulation of the airflow.

In fact, an airfoil with camber will produce lift even at zero angle between the chord line and the oncoming air ("angle of attack"). For a cambered airfoil there is some negative angle at which no lift is produced, the "angle

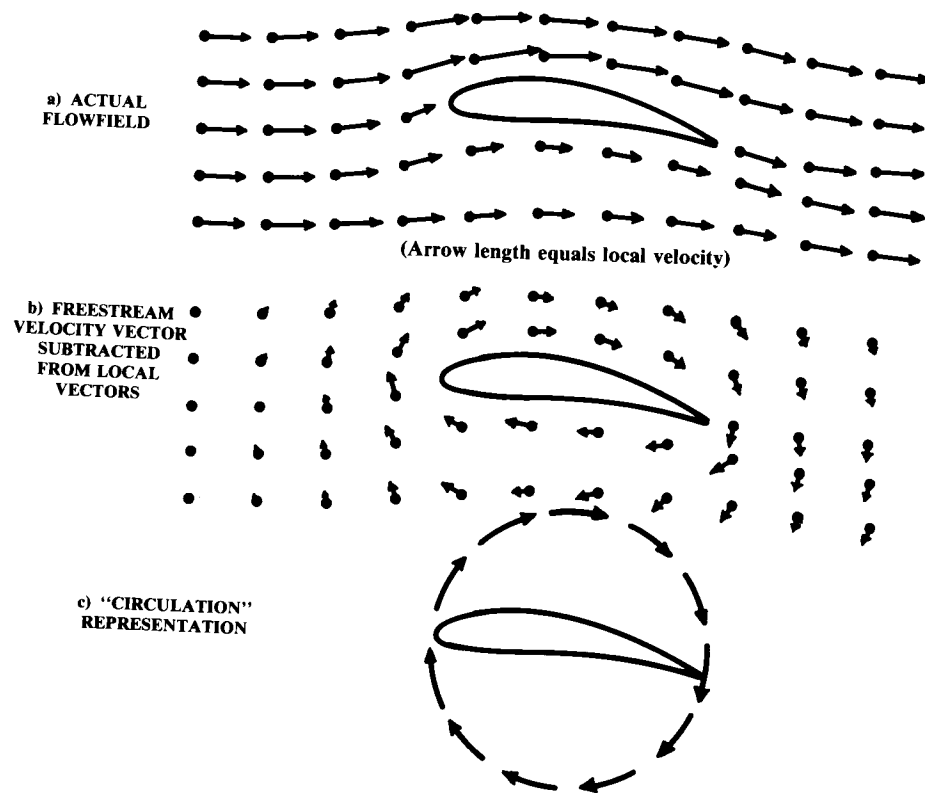


Fig. 4.3 Airfoil flowfield and circulation.

of zero lift." This negative angle is approximately equal (in degrees) to the percent camber of the airfoil.

Odd as it sounds, an airfoil in two-dimensional (2-D) flow does not experience any drag due to the creation of lift. The pressure forces produced in the generation of lift are at right angles to the oncoming air. All two-dimensional airfoil drag is produced by skin friction and pressure effects resulting from flow separation and shocks. It is only in three-dimensional (3-D) flow that drag due to lift is produced.

The airfoil section lift, drag, and pitching moment are defined in non-dimensional form in Eqs. (4.1), (4.2), and (4.3). By definition, the lift force is perpendicular to the flight direction while the drag force is parallel to the flight direction. The pitching moment is usually negative when measured about the aerodynamic center, implying a nose-down moment. Note that 2-D airfoil characteristics are denoted by lowercase subscripts (i.e., C_l) whereas the 3-D wing characteristics are denoted by uppercase subscripts (i.e., C_L).

$$C_l = \frac{\text{section lift}}{qc} \quad (4.1)$$

$$C_d = \frac{\text{section drag}}{qc} \quad (4.2)$$

$$C_m = \frac{\text{section moment}}{qc^2} \quad (4.3)$$

where

c = chord length

q = dynamic pressure = $\rho V^2/2$

α = angle of attack

C_{l_α} = slope of the lift curve = 2π (typically)

The point about which the pitching moment remains constant for any angle of attack is called the "aerodynamic center." The aerodynamic center is not the same as the airfoil's center of pressure (or lift). The center of pressure is usually behind the aerodynamic center. The location of the center of pressure varies with angle of attack for most airfoils.

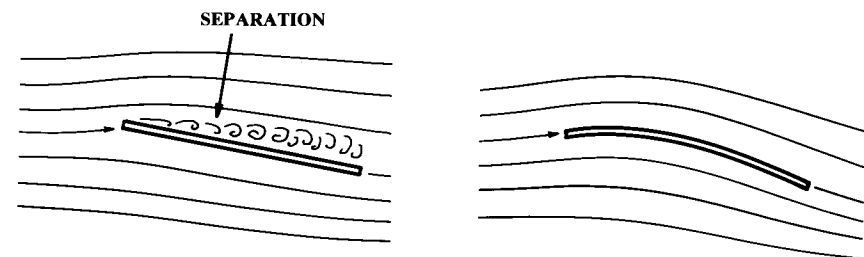


Fig. 4.4 Effect of camber on separation.

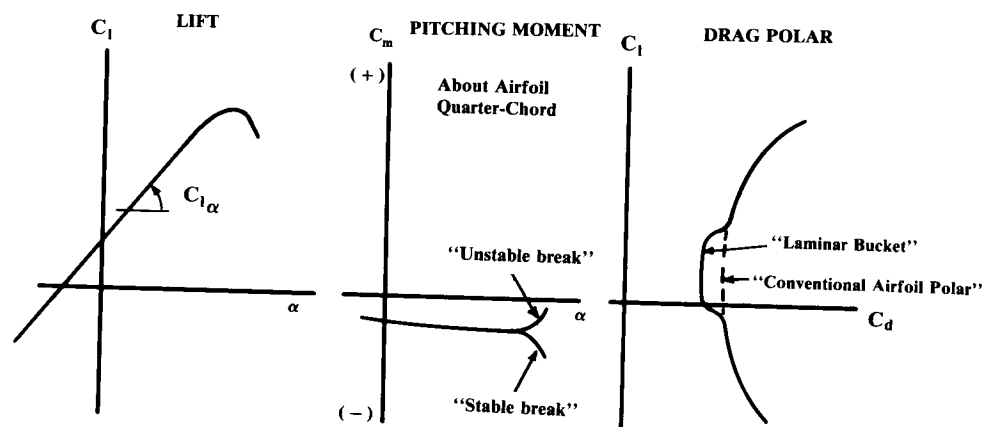


Fig. 4.5 Airfoil lift, drag, and pitching moment.

Pitching moment is measured about some reference point, typically the quarter-chord point (25% of the chord length back from the leading edge). The pitching moment is almost independent of angle of attack about the quarter-chord for most airfoils at subsonic speeds (i.e., the aerodynamic center is usually at the quarter-chord point.)

Lift, drag, and pitching-moment characteristics for a typical airfoil are shown in Fig. 4.5.

Airfoil characteristics are strongly affected by the "Reynolds number" at which they are operating. Reynolds number, the ratio between the dynamic and the viscous forces in a fluid, is equal to $(\rho V l / \mu)$, where V is the velocity, l the length the fluid has traveled down the surface, ρ the fluid density, and μ the fluid viscosity coefficient. The Reynolds number influences whether the flow will be laminar or turbulent, and whether flow separation will occur. A typical aircraft wing operates at a Reynolds number of about ten million.

Figure 4.5 illustrates the so-called "laminar bucket." For a "laminar" airfoil operating at the design Reynolds number there is a range of lift coefficient for which the flow remains laminar over a substantial part of the airfoil. This causes a significant reduction of drag for a given lift coefficient.

However, this effect is very dependent upon the Reynolds number as well as the actual surface smoothness. For example, dirt, rain, or insect debris on the leading edge may cause the flow to become turbulent, causing an increase in drag to the dotted line shown in Fig. 4.5. This also can change the lift and pitching-moment characteristics.

In several canarded homebuilt designs with laminar airfoils, entering a light rainfall will cause the canard's airflow to become turbulent, reducing canard lift and causing the aircraft to pitch downward. Earlier, nonlaminar airfoils were designed assuming turbulent flow at all times and do not experience this effect.

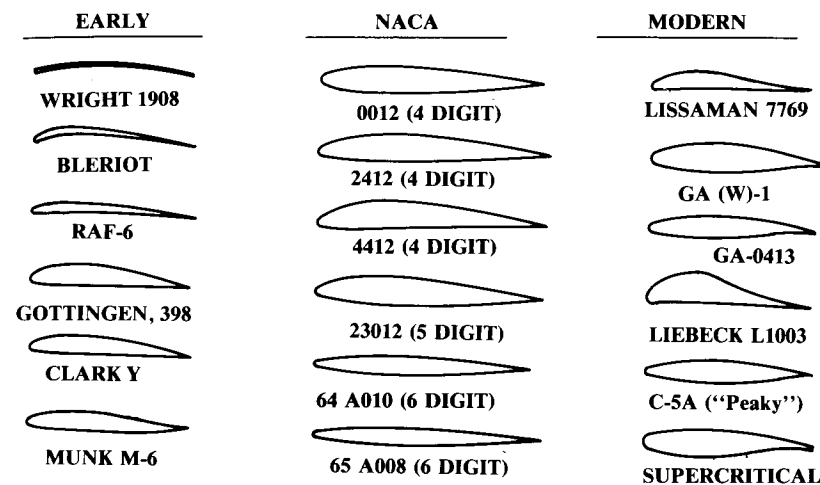


Fig. 4.6 Typical airfoils.

Airfoil Families

A variety of airfoils is shown in Fig. 4.6. The early airfoils were developed mostly by trial and error. In the 1930's, the NACA developed a widely-used family of mathematically defined airfoils called the "four-digit" airfoils. In these, the first digit defined the percent camber, the second defined the location of the maximum camber, and the last two digits defined the airfoil maximum thickness in percent of chord. While rarely used for wing design today, the uncambered four-digit airfoils are still commonly used for tail surfaces of subsonic aircraft.

The NACA five-digit airfoils were developed to allow shifting the position of maximum camber forward for greater maximum lift. The six-series airfoils were designed for increased laminar flow, and hence reduced drag. Six-series airfoils such as the 64A series are still widely used as a starting point for high-speed-wing design. The Mach 2 F-15 fighter uses the 64A airfoil modified with camber at the leading edge. Geometry and characteristics of these "classical" airfoils are summarized in Ref. 2, a must for every designer's library.

Airfoil Design

In the past, the designer would select an airfoil (or airfoils) from such a catalog. This selection would consider factors such as the airfoil drag during cruise, stall and pitching-moment characteristics, the thickness available for structure and fuel, and the ease of manufacture. With today's computational airfoil design capabilities, it is becoming common for the airfoil shapes for a wing to be custom-designed.

Modern airfoil design is based upon inverse computational solutions for desired pressure (or velocity) distributions on the airfoil. Methods have been

developed for designing an airfoil such that the pressure differential between the top and bottom of the airfoil quickly reaches a maximum value attainable without airflow separation. Toward the rear of the airfoil, various pressure recovery schemes are employed to prevent separation near the trailing edge.

These airfoil optimization techniques result in airfoils with substantial pressure differentials (lift) over a much greater percent of chord than a classical airfoil. This permits a reduced wing area (and wetted area) for a required amount of lift. Such airfoil design methods go well beyond the scope of this book.

Another consideration in modern airfoil design is the desire to maintain laminar flow over the greatest possible part of the airfoil. Laminar flow can be maintained by providing a negative pressure gradient, i.e., by having the pressure continuously drop from the leading edge to a position close to the trailing edge. This tends to "suck" the flow rearward, promoting laminar flow.

A good laminar-flow airfoil combined with smooth fabrication methods can produce a wing with laminar flow over about 50–70% of the wing. Figure 4.7 shows a typical laminar flow airfoil and its pressure distribution.

As an airfoil generates lift the velocity of the air passing over its upper surface is increased. If the airplane is flying at just under the speed of sound, the faster air traveling over the upper surface will reach supersonic speeds causing a shock to exist on the upper surface, as shown in Fig. 4.8. The speed at which supersonic flow first appears on the airfoil is called the "Critical Mach Number" (M_{crit}).

The upper-surface shock creates a large increase in drag, along with a reduction in lift and a change in the pitching moment. The drag increase

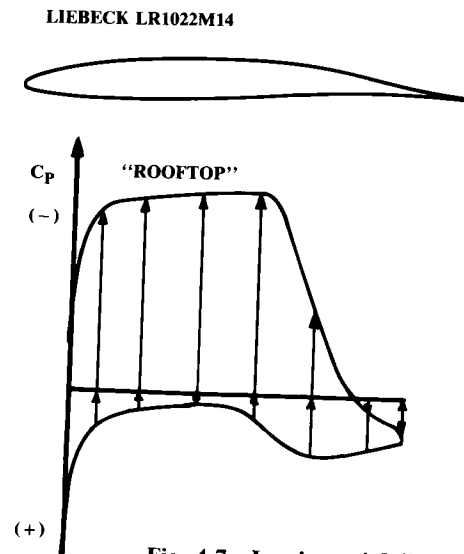


Fig. 4.7 Laminar airfoil.

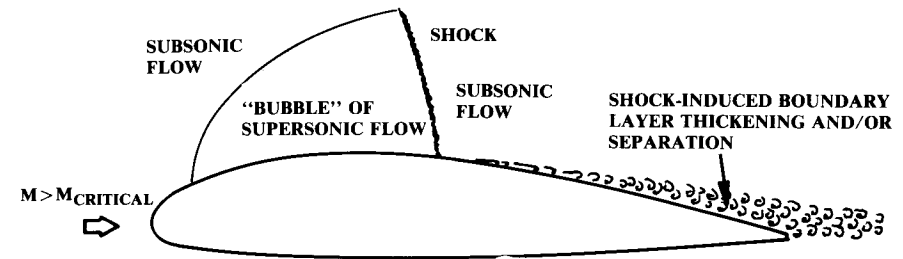


Fig. 4.8 Transonic effects.

comes from the tendency of the rapid pressure rise across the shock to thicken or even separate the boundary layer.

A "supercritical" airfoil is one designed to minimize these effects. Modern computational methods allow design of airfoils in which the upper-surface shock is minimized or even eliminated by spreading the lift in the chordwise direction, thus reducing the upper surface velocity for a required total lift. This increases the Critical Mach Number.

Design Lift Coefficient

For early conceptual design work, the designer must frequently rely upon existing airfoils. From existing airfoils, the one should be selected that comes closest to having the desired characteristics.

The first consideration in initial airfoil selection is the "design lift coefficient." This is the lift coefficient at which the airfoil has the best L/D (shown in Fig. 4.9 as the point on the airfoil drag polar that is tangent to a line from the origin and closest to the vertical axis).

In subsonic flight a well-designed airfoil operating at its design lift coefficient has a drag coefficient that is little more than skin-friction drag. The aircraft should be designed so that it flies the design mission at or near the design lift coefficient to maximize the aerodynamic efficiency.

As a first approximation, it can be assumed that the wing lift coefficient, C_L , equals the airfoil lift coefficient, C_l . In level flight the lift must equal the weight, so the required design lift coefficient can be found as follows:

$$W = L = qSC_L \approx qSC_l \quad (4.4)$$

$$C_l = \frac{1}{q} \left(\frac{W}{S} \right) \quad (4.5)$$

Dynamic pressure (q) is a function of velocity and altitude. By assuming a wing loading (W/S) as described later, the design lift coefficient can be calculated for the velocity and altitude of the design mission.

Note that the actual wing loading will decrease during the mission as fuel is burned. Thus, to stay at the design lift coefficient, the dynamic pressure must be steadily reduced during the mission by either slowing down, which

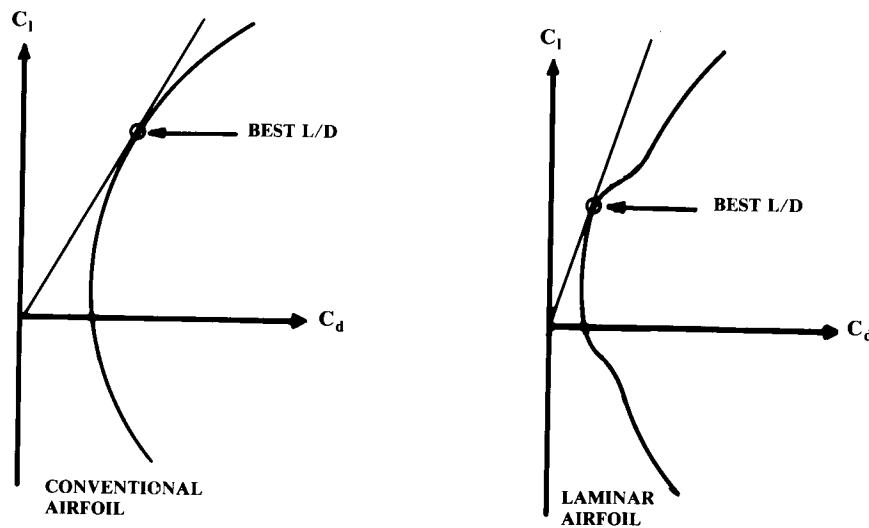


Fig. 4.9 Design lift coefficient.

is undesirable, or climbing to a higher altitude. This explains the “cruise-climb” followed by an aircraft trying to maximize range.

In actual practice, a design lift coefficient usually will be based upon past experience, and for most types of aircraft typically will be around 0.5. In fact, the initial selection of the airfoil is often simply based upon prior experience or copied from some successful design.

Stall

Stall characteristics play an important role in airfoil selection. Some airfoils exhibit a gradual reduction in lift during a stall, while others show a violent loss of lift, accompanied by a rapid change in pitching moment. This difference reflects the existence of three entirely different types of airfoil stall.

“Fat” airfoils (round leading edge and t/c greater than about 14%) stall from the trailing edge. The turbulent boundary layer increases with angle of attack. At around 10 deg the boundary layer begins to separate, starting at the trailing edge and moving forward as the angle of attack is further increased. The loss of lift is gradual. The pitching moment changes only a small amount.

Thinner airfoils stall from the leading edge. If the airfoil is of moderate thickness (about 6–14%), the flow separates near the nose at a very small angle of attack, but immediately reattaches itself so that little effect is felt. At some higher angle of attack the flow fails to reattach, which almost immediately stalls the entire airfoil. This causes an abrupt change in lift and pitching moment.

Very thin airfoils exhibit another form of stall. As before, the flow separates from the nose at a small angle of attack and reattaches almost

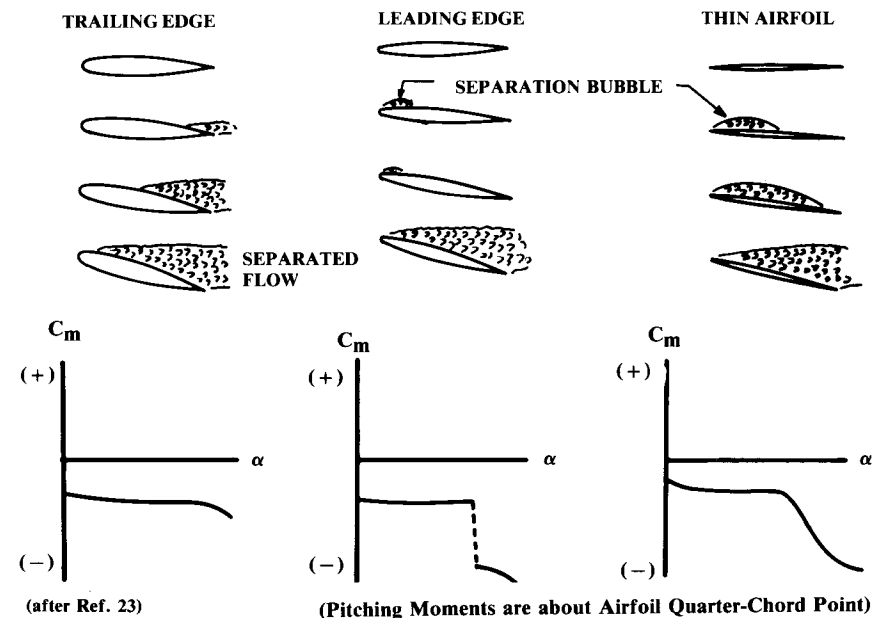


Fig. 4.10 Types of stall.

immediately. However, for a very thin airfoil this “bubble” continues to stretch towards the trailing edge as the angle of attack is increased. At the angle of attack where the bubble stretches all the way to the trailing edge, the airfoil reaches its maximum lift. Beyond that angle of attack, the flow is separated over the whole airfoil, so the stall occurs. The loss of lift is smooth, but large changes in pitching moment are experienced. The three types of stall characteristics are depicted in Fig. 4.10.

Twisting the wing such that the tip airfoils have a reduced angle of attack compared to the root (“washout”) can cause the wing to stall first at the root. This provides a gradual stall even for a wing with a poorly stalling airfoil. Also, the turbulent wake off the stalled wingroot will vibrate the horizontal tail, notifying the pilot that a stall is imminent.

In a similar fashion, the designer may elect to use different airfoils at the root and tip, with a tip airfoil selected which stalls at a higher angle of attack than the root airfoil. This provides good flow over the ailerons for roll control at an angle of attack where the root is stalled.

If different airfoils are used at the root and tip, the designer must develop the intermediate airfoils by interpolation (discussed later). These intermediate airfoils will have section characteristics somewhere between those of the root and tip airfoils, and can also be estimated by interpolation. This interpolation of section characteristics does not work for modern supercritical or laminar-flow airfoils. Estimation of the section characteristics in those cases must be done computationally.

Stall characteristics for thinner airfoils can be improved with various leading-edge devices such as slots, slats, leading-edge flaps, Krueger flaps,

and active methods (e.g., suction or blowing). These are discussed in the Aerodynamics Chapter.

Wing stall is directly related to airfoil stall only for high-aspect-ratio, unswept wings. For lower aspect ratio or highly swept wings the 3-D effects dominate stall characteristics, and airfoil stall characteristics can be essentially ignored in airfoil selection.

Pitching moment must also be considered in airfoil selection. Horizontal tail or canard size is directly affected by the magnitude of the wing pitching moment to be balanced. Some of the supercritical airfoils use what is called "rear-loading" to increase lift without increasing the region of supersonic flow. This produces an excellent L/D , but can cause a large nose-down pitching moment. If this requires an excessive tail area, the total aircraft drag may be increased, not reduced.

For a stable tailless or flying-wing aircraft, the pitching moment must be near zero. This usually requires an "S"-shaped camber with the characteristic upward reflex at the trailing edge. Reflexed airfoils have poorer L/D than an airfoil designed without this constraint. This tends to reduce some of the benefit that flying wings experience due to their reduced wetted area. However, a computerized, "active" flight control system can remove the requirement for natural stability, and thus allow a nonreflexed airfoil.

Airfoil Thickness Ratio

Airfoil thickness ratio has a direct effect on drag, maximum lift, stall characteristics, and structural weight. Figure 4.11 illustrates the effect of thickness ratio on subsonic drag. The drag increases with increasing thickness due to increased separation.

Figure 4.12 shows the impact of thickness ratio on Critical Mach Number, the Mach number at which supersonic flow first appears over the wing. A supercritical airfoil tends to minimize shock formation and can be used to reduce drag for a given thickness ratio or to permit a thicker airfoil at the same drag level.

The thickness ratio affects the maximum lift and stall characteristics primarily by its effect on the nose shape. For a wing of fairly high aspect ratio and moderate sweep, a larger nose radius provides a higher stall angle and a greater maximum lift coefficient, as shown in Fig. 4.13.

The reverse is true for low-aspect-ratio, swept wings, such as a delta wing. Here, a sharper leading edge provides greater maximum lift due to the formation of vortices just behind the leading edge. These leading edge vortices act to delay wing stall. This 3-D effect is discussed in the Aerodynamics Chapter.

Thickness also affects the structural weight of the wing. Statistical equations for wing weight show that the wing structural weight varies approximately inversely with the square root of the thickness ratio. Halving the thickness ratio will increase wing weight by about 41%. The wing is typically about 15% of the total empty weight, so halving the thickness ratio would increase empty weight by about 6%. When applied to the sizing equation, this can have a major impact.

For initial selection of the thickness ratio, the historical trend shown in Fig. 4.14 can be used. Note that a supercritical airfoil would tend to be about 10% thicker (i.e., conventional airfoil thickness ratio times 1.1) than the historical trend.

Frequently the thickness is varied from root to tip. Due to fuselage effects, the root airfoil of a subsonic aircraft can be as much as 20–60% thicker than the tip airfoil without greatly affecting the drag. This is very beneficial, resulting in a structural weight reduction as well as more volume for fuel and landing gear. This thicker root airfoil should extend to no more than about 30% of the span.

Other Airfoil Considerations

Another important aspect of airfoil selection is the intended Reynolds number. Each airfoil is designed for a certain Reynolds number. Use of an airfoil at a greatly different Reynolds number (half an order of magnitude or so) can produce section characteristics much different from those expected.

This is especially true for the laminar-flow airfoils, and is most crucial when an airfoil is operated at a lower than design Reynolds number. In the past this has been a problem for homebuilt and sailplane designers, but there are now suitable airfoils designed especially for these lower Reynolds number aircraft.

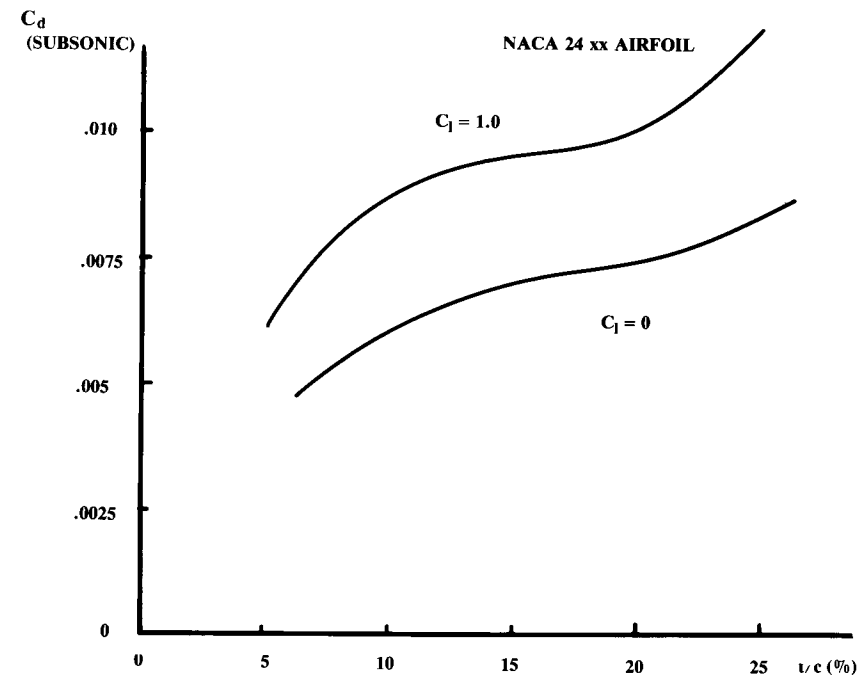


Fig. 4.11 Effects of t/c on drag.

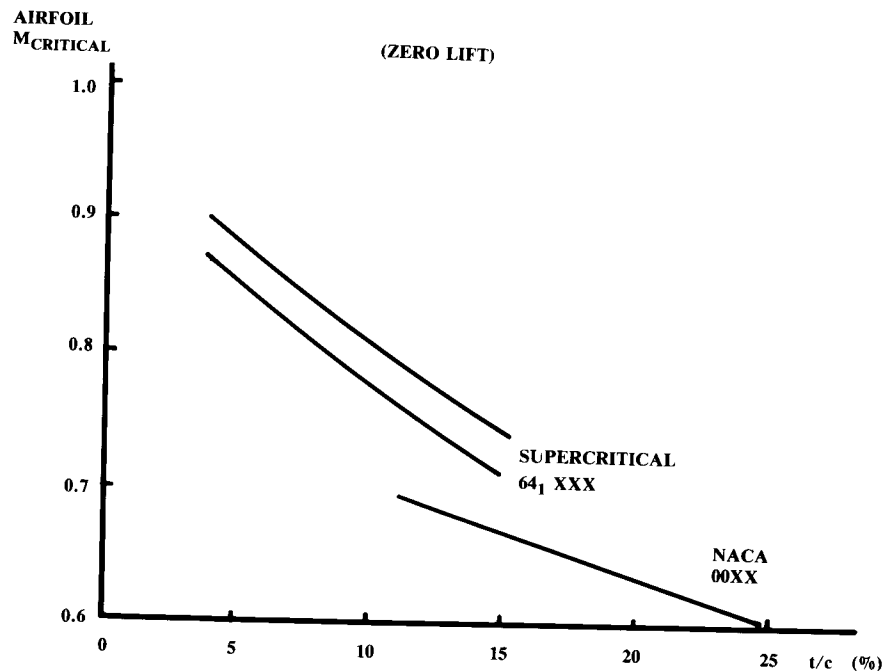
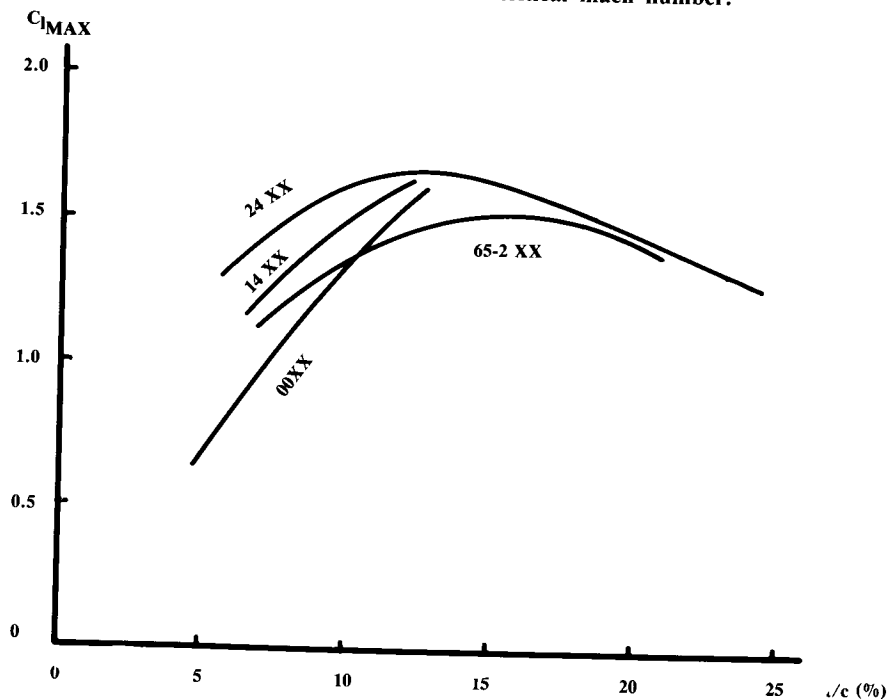
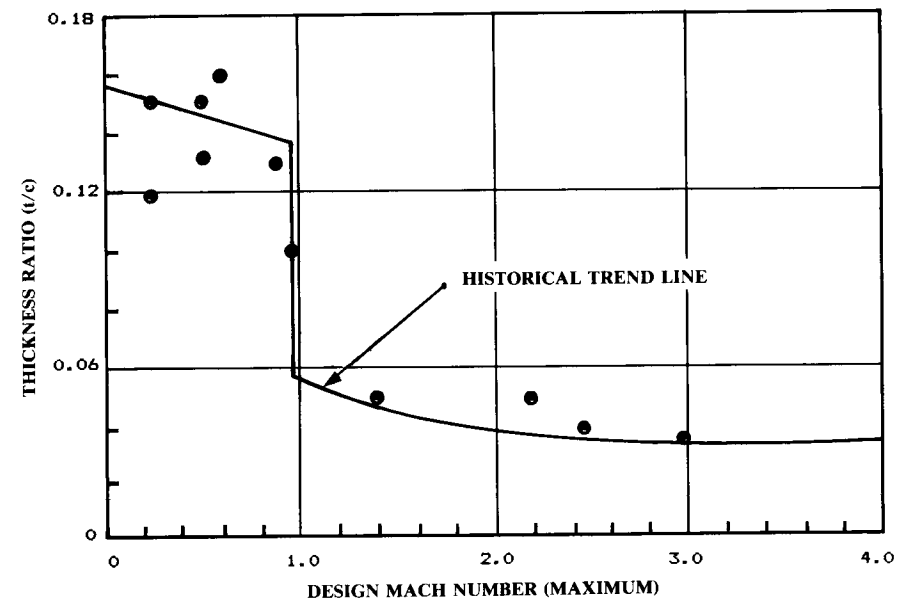
Fig. 4.12 Effect of t/c on critical mach number.Fig. 4.13 Effect of t/c on maximum lift.

Fig. 4.14 Thickness ratio historical trend.

The laminar airfoils require extremely smooth skins as well as exact control over the actual, as-manufactured shape. These can drive the cost up significantly. Also, the camouflage paints used on military aircraft are rough compared to bare metal or composite skins. This must be considered before selecting certain airfoils.

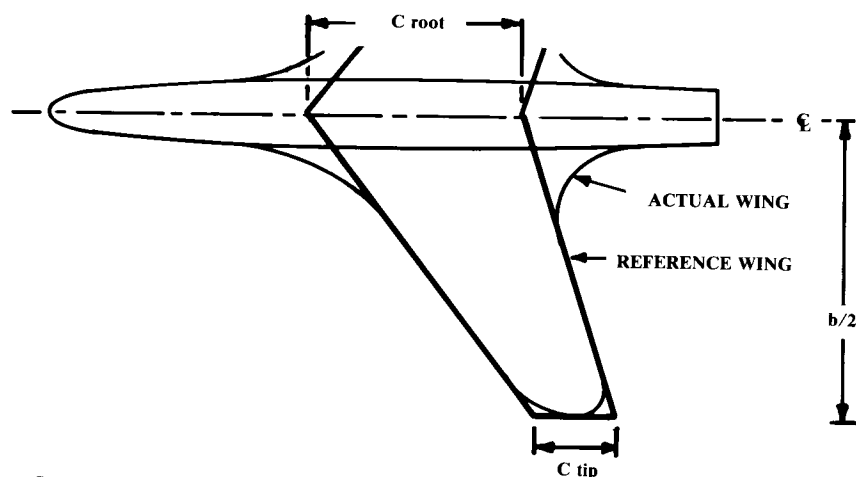
While an understanding of the factors important to airfoil selection is important, an aircraft designer should not spend too much time trying to pick exactly the "right" airfoil in early conceptual design. Later trade studies and analytical design tools will determine the desired airfoil characteristics and geometry. For early conceptual layout, the selected airfoil is important mostly for determining the thickness available for structure, landing gear, and fuel.

Appendix A.3 provides geometry and section characteristics for a few airfoils useful in conceptual design. For swept-wing supersonic aircraft, the NACA 64A and 65A sections are good airfoils for initial design. The appendix describes a supercritical section suitable for transports and other high-subsonic aircraft, along with a typical modern NASA section for general aviation. A few specialized airfoils are provided for other applications.

The airfoils presented in Appendix A.3 are not being recommended as the "best" sections for those applications, but rather as reasonable airfoils with which to start a conceptual design. Again, Ref. 2 is highly recommended.

4.3 WING GEOMETRY

The "reference" ("trapezoidal") wing is the basic wing geometry used to begin the layout. Figures 4.15 and 4.16 show the key geometric parameters of the reference wing.

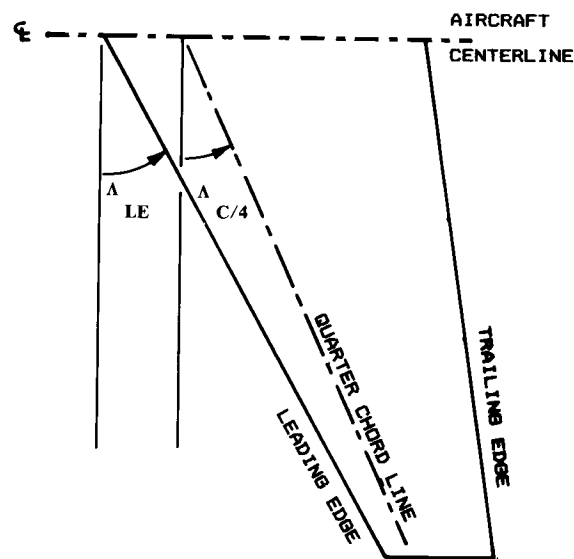


- S = REFERENCE WING AREA
 C = CHORD (DISTANCE L.E. TO T.E.)
 A = ASPECT RATIO = b^2/S
 t/c = AIRFOIL THICKNESS RATIO (MAXIMUM THICKNESS/CHORD)
 λ = TAPER RATIO = C_{tip}/C_{root}
 b = SPAN

GIVEN: W/S , A , λ

$$S = W/(W/S) \quad b = \sqrt{A \cdot S} \quad C_{root} = 2 \cdot S/[b(1 + \lambda)] \quad C_{tip} = \lambda \cdot C_{root}$$

Fig. 4.15 Wing geometry.



$$\tan \Lambda_{LE} = \tan \Lambda_{C/4} + [(1 - \lambda)/A(1 + \lambda)]$$

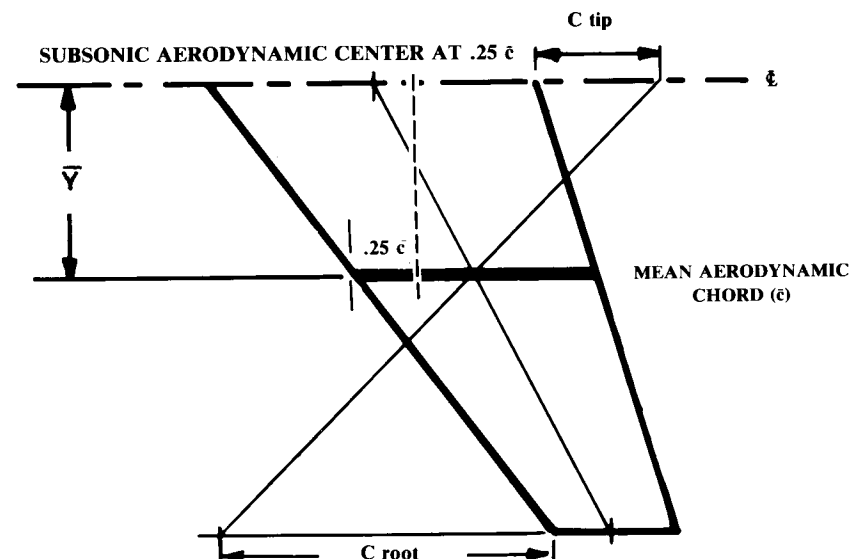
Fig. 4.16 Wing sweep Λ .

Note that the reference wing is fictitious, and extends through the fuselage to the aircraft centerline. Thus the reference wing area includes the part of the reference wing which sticks into the fuselage. For the reference wing, the root airfoil is the airfoil of the trapezoidal reference wing at the centerline of the aircraft, not where the actual wing connects to the fuselage.

There are two key sweep angles, as shown in Fig. 4.16. The leading-edge sweep is the angle of concern in supersonic flight. To reduce drag it is common to sweep the leading edge behind the Mach cone. The sweep of the quarter-chord line is the sweep most related to subsonic flight. It is important to avoid confusing these two sweep angles. The equation at the bottom of Fig. 4.16 allows converting from one sweep angle to the other.

Airfoil pitching moment data in subsonic flow is generally provided about the quarter-chord point, where the airfoil pitching moment is essentially constant with changing angle of attack (i.e., the "aerodynamic center"). In a similar fashion, such a point is defined for the complete trapezoidal wing and is based on the concept of the "mean aerodynamic chord." The mean aerodynamic chord (Fig. 4.17) is the chord " $\bar{c}$ " of an airfoil, located at some distance " $\bar{Y}$ " from the centerline.

The entire wing has its mean aerodynamic center at approximately the same percent location of the mean aerodynamic chord as that of the airfoil



$$\bar{c} = (2/3) C_{root} (1 + \lambda + \lambda^2)/(1 + \lambda)$$

$$\bar{Y} = (b/6)[(1 + 2\lambda)(1 + \lambda)]$$

TYPICAL, WING AERODYNAMIC CENTER = $.25 \bar{c}$ SUBSONIC
 = $.4 \bar{c}$ SUPERSONIC

Fig. 4.17 Mean aerodynamic chord.

alone. In subsonic flow, this is at the quarter-chord point on the mean aerodynamic chord. In supersonic flow, the aerodynamic center moves back to about 40% of the mean aerodynamic chord. The designer uses the mean aerodynamic chord and the resulting aerodynamic center point to position the wing properly. Also, the mean aerodynamic chord will be important to stability calculations. Figure 4.17 illustrates a graphical method for finding the mean aerodynamic chord of a trapezoidal-wing planform.

The required reference wing area ("S") can be determined only after the takeoff gross weight is determined. The shape of the reference wing is determined by its aspect ratio, taper ratio, and sweep.

Aspect Ratio

The first to investigate aspect ratio in detail were the Wright Brothers, using a wind tunnel they constructed. They found that a long, skinny wing (high aspect ratio) has less drag for a given lift than a short, fat wing (low aspect ratio). This is due to the 3-D effects.

As most early wings were rectangular in shape, the aspect ratio was initially defined as simply the span divided by the chord. For a tapered wing, the aspect ratio is defined as the span squared divided by the area (which defaults to the earlier definition for a wing with no taper).

When a wing is generating lift, it has a reduced pressure on the upper surface and an increased pressure on the lower surface. The air would like to "escape" from the bottom of the wing, moving to the top. This is not possible in 2-D flow unless the airfoil is leaky (a real problem with some fabric wing materials unless properly treated). However, for a real, 3-D wing, the air can escape around the wing tip.

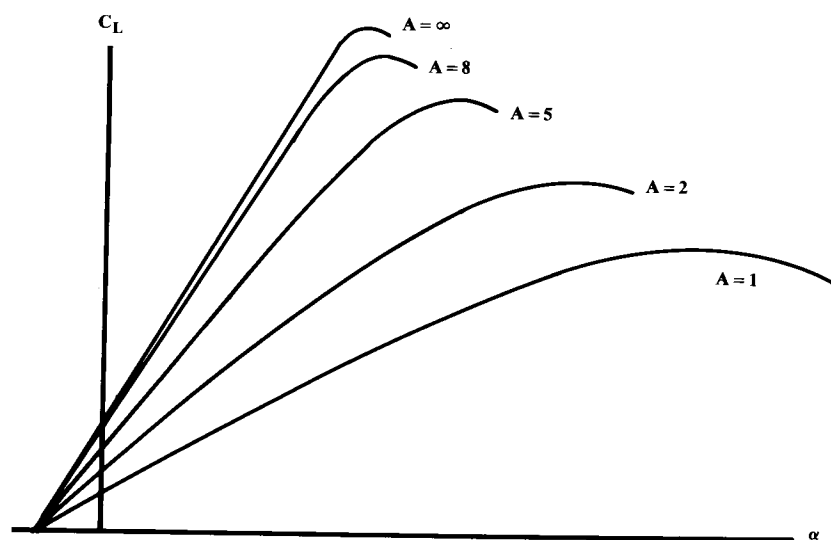


Fig. 4.18 Effect of aspect ratio on lift.

Air escaping around the wing tip lowers the pressure difference between the upper and the lower surfaces. This reduces lift near the tip. Also, the air flowing around the tip flows in a circular path when seen from the front, and in effect pushes down on the wing. Strongest near the tip, this reduces the effective angle of attack of the wing airfoils. This circular, or "vortex" flow pattern continues downstream behind the wing.

A wing with a high aspect ratio has tips farther apart than an equal area wing with a low aspect ratio. Therefore, the amount of the wing affected by the tip vortex is less for a high aspect ratio wing than for a low-aspect-ratio wing, and the strength of the tip vortex is reduced. Thus, the high-aspect-ratio wing does not experience as much of a loss of lift and increase of drag due to tip effects as a low-aspect-ratio wing of equal area.

(It is actually the wing span which determines the drag due to lift. However, wing area is usually held constant unless widely different aircraft concepts are being evaluated. When wing area is held constant, the wing span varies as the square root of the aspect ratio.)

As was shown in Fig. 3.6, the maximum subsonic L/D of an aircraft increases approximately by the square root of an increase in aspect ratio (when wing area and S_{wet}/S_{ref} are held constant). On the other hand, the wing weight also increases with increasing aspect ratio, by about the same factor.

Another effect of changing aspect ratio is a change in stalling angle. Due to the reduced effective angle of attack at the tips, a lower-aspect-ratio wing will stall at a higher angle of attack than a higher-aspect-ratio wing (Fig. 4.18). This is one reason why tails tend to be of lower aspect ratio. Delaying tail stall until well after the wing stalls assures adequate control.

Table 4.1 Aspect ratio

Sailplane equivalent* aspect ratio = 4.464 (best L/D). ⁶⁹		
Propeller aircraft	Equivalent aspect ratio	
Homebuilt	6.0	
General aviation—single engine	7.6	
General aviation—twin engine	7.8	
Agricultural aircraft	7.5	
Twin turboprop	9.2	
Flying boat	8.0	
Jet aircraft	Equivalent aspect Ratio = aM_{max}^C	
	a	C
Jet trainer	4.737	− 0.979
Jet fighter (dogfighter)	5.416	− 0.622
Jet fighter (other)	4.110	− 0.622
Military cargo/bomber	5.570	− 1.075
Jet transport	7.50	0

*Equivalent aspect ratio = wing span squared/(wing and canard areas)

Conversely, a canard can be made to stall before the wing by making it a very high aspect ratio surface. This prevents the pilot from stalling the wing, and is seen in several canarded homebuilt designs.

Later in the design process, the aspect ratio will be determined by a trade study in which the aerodynamic advantages of a higher aspect ratio are balanced against the increased weight. For initial wing layout, the values and equations provided in Table 4.1 can be used. These were determined through statistical analysis of a number of aircraft, using data from Ref. 1.

Sailplane aspect ratio was found to be directly related to the desired glide ratio, which equals the L/D . Propeller aircraft showed no clear statistical trend, so average values are presented. Jet aircraft evidence a strong trend of aspect ratio decreasing with increasing Mach number evidence due to drag due to lift becoming relatively less important at higher speeds. Designers of high-speed aircraft thus use lower-aspect-ratio wings to save weight.

Note that, for statistical purposes, Table 4.1 uses an equivalent wing area that includes the canard area in defining the aspect ratio of an aircraft with a lifting canard. To determine the actual wing geometric aspect ratio, it is necessary to decide how to split the lifting area between the wing and canard. Typically, the canard will have about 10–25% of the total lifting area, so the wing aspect ratio becomes the statistically determined aspect ratio divided by 0.9–0.75.

Wing Sweep

Wing sweep is used primarily to reduce the adverse effects of transonic and supersonic flow. Theoretically, shock formation on a swept wing is determined not by the actual velocity of the air passing over the wing, but rather by the air velocity in a direction perpendicular to the leading edge of the wing. This result, first applied by the Germans during World War II, allows an increase in Critical Mach Number by the use of sweep.

At supersonic speeds the loss of lift associated with supersonic flow can be reduced by sweeping the wing leading edge aft of the Mach cone angle [$\arcsin(1/\text{Mach No.})$].

Figure 4.19 shows a historical trend line for wing leading-edge sweep vs Mach number. Note that sweep is defined aft of a line perpendicular to the flight direction, while the Mach angle is defined with respect to the flight direction. Thus, the line labeled “ $90 - \arcsin(1/\text{Mach No.})$ ” is the wing sweep required to place the wing leading edge exactly on the Mach cone.

The historical trend differs from this theoretical result for two reasons. In the high-speed range, it becomes structurally impractical to sweep the wing past the Mach cone. In this speed regime, over about Mach 2.5, it is necessary to use sharp or nearly sharp airfoils.

Selecting the wing sweep to equal the Mach-cone angle would indicate a zero sweep for speeds at or below Mach 1.0. However, in the transonic speed regime (roughly Mach 0.9–1.2) the desire for a high critical Mach number predominates. This requires subsonic airflow velocity over the airfoil (when measured perpendicular to the leading edge), and thus a swept wing.

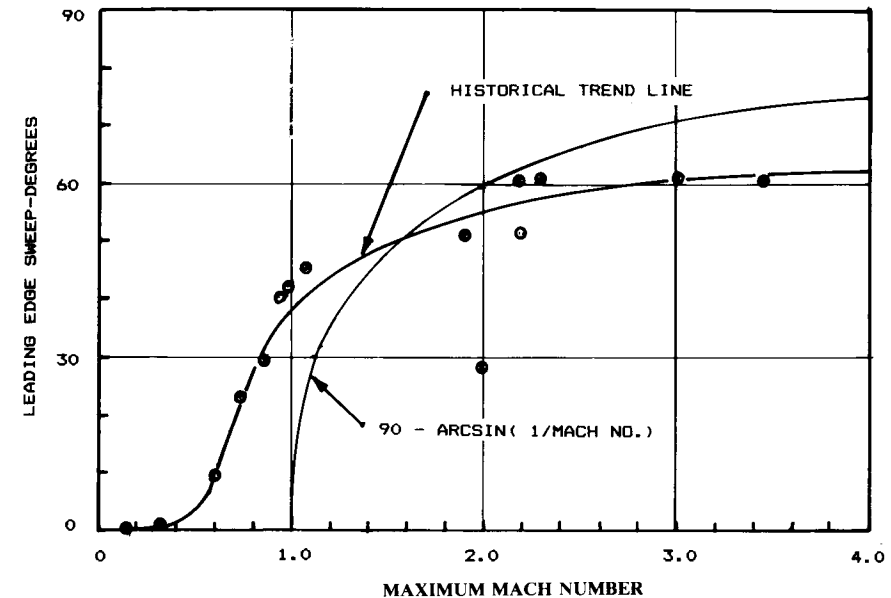


Fig. 4.19 Wing sweep historical trend.

The exact wing sweep required to provide the desired Critical Mach Number depends upon the selected airfoil(s), thickness ratio, taper ratio, and other factors. For initial wing layout the trend line of Fig. 4.19 is reasonable.

There is no theoretical difference between sweeping a wing aft and sweeping it forward. In the past, wings have been swept aft because of the structural divergence problem associated with forward sweep. With the use of composite materials, this can be avoided for a small weight penalty.

Also, there is no reason why one cannot sweep one wing aft and the other wing forward, creating an “oblique wing.” This arrangement produces unusual control responses, but a computerized flight control system can easily provide normal handling qualities. The oblique wing also tends to have lower wave drag.

There are other reasons for sweeping a wing. For example, the fuselage layout may not otherwise allow locating the wing carry-through structure at the correct place for balancing the aircraft. Canarded aircraft with pusher engines are frequently tail-heavy, requiring wing sweep to move the aerodynamic center back far enough for balance. This is why most canard pushers have swept wings.

Wing sweep improves stability. A swept wing has a natural dihedral effect. In fact, it is frequently necessary to use zero or negative dihedral on a swept wing to avoid excessive stability.

If an aircraft has its vertical tails at the wingtips, sweeping the wing will push the tails aft, increasing their effectiveness. This is also seen on many canard pusher aircraft.

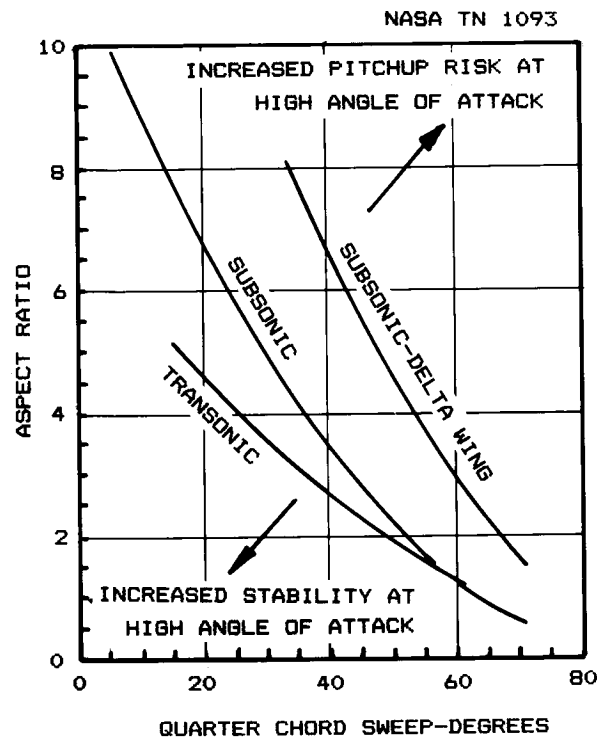


Fig. 4.20 Tail-off pitchup boundaries.

Note on Fig. 4.19 the data point at Mach 2.0 and leading-edge sweep just under 30 deg. This is the Lockheed F-104, which used a different approach for reducing drag at supersonic speeds. The F-104 had a razor-sharp leading edge, so sharp that it was covered on the ground for the safety of line personnel. The F-104 also had a very thin wing, only 3.4% thick.

The wing sweep and aspect ratio together have a strong effect on the wing-alone pitchup characteristics. "Pitchup" is the highly undesirable tendency of some aircraft, upon reaching an angle of attack near stall, to suddenly and uncontrollably increase the angle of attack. The aircraft continues pitching up until it stalls and departs totally out of control. The F-16 fighter requires a computerized angle-of-attack limiter to prevent a severe pitchup problem at about 25-deg angle of attack.

Figure 4.20 describes boundaries for pitchup avoidance for combinations of wing quarter-chord sweep angle and aspect ratio. Pitchup avoidance should be considered for military fighters, aerobatic aircraft, general-aviation aircraft, and trainers.

These boundaries may limit the allowable aspect ratio to a value less than that estimated earlier. However, Fig. 4.20 provides data for the wing alone. If a properly designed horizontal tail is used, the aspect ratio may be higher than that allowed by the graph. This is discussed later. Also, a large, all-

moving canard such as that seen on the Grumman X-29 can be used to control a pitchup tendency. However, this requires a computerized flight control system.

For high-speed flight, a swept wing is desirable. For cruise as well as takeoff and landing, an unswept wing is desirable. A wing of variable sweep would offer the best of both worlds. Variable sweep was first flight-tested in the 1950's, and is now on several operational military aircraft including the F-111, F-14, B-1B, and the European Tornado and Soviet Backfire.

For design purposes, the planform for a variable-sweep aircraft should be developed in the unswept position, and then swept to the desired leading-edge angle for high-speed flight. The pivot position about which the wing is swept must be near the thickest part of the chord, between about the 30- and 40%-chord locations. Also, provisions must be made for smoothly fairing the wing root in both extended and fully-swept positions.

Controlling the balance of a variable sweep aircraft is a major design problem. When the wing swings aft, the aerodynamic center moves with it. The center of gravity also moves due to the wing movement, but not nearly as much as the aerodynamic center. To balance the aircraft, either fuel must be pumped to move the center of gravity, or the tail must provide a tremendous down-load (or both).

Yet another problem with the variable-sweep wing is the weight penalty associated with the pivot mechanism and less-than-optimal load paths. As shown in Table 3.1, variable sweep increases total empty weight roughly 4%. The detailed statistical weight equations of Chapter 15 show a 19% increase in the weight of the wing itself if it has variable sweep.

Taper Ratio

Wing taper ratio, λ , is the ratio between the tip chord and the centerline root chord. Most wings of low sweep have a taper ratio of about 0.4–0.5. Most swept wings have a taper ratio of about 0.2–0.3.

Taper affects the distribution of lift along the span of the wing. As proven by the Prandtl wing theory early in this century, minimum drag due to lift, or "induced" drag, occurs when the lift is distributed in an elliptical fashion. For an untwisted and unswept wing, this occurs when the wing plan-

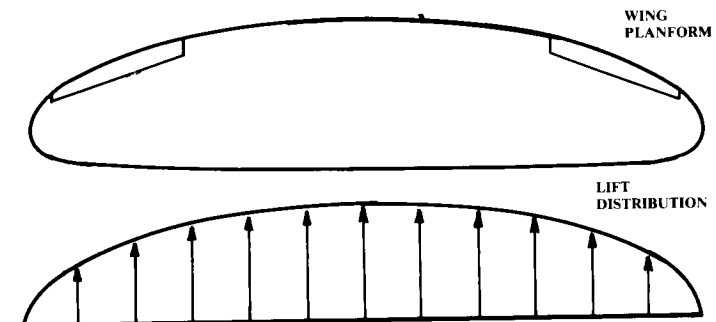


Fig. 4.21 Elliptical wing.

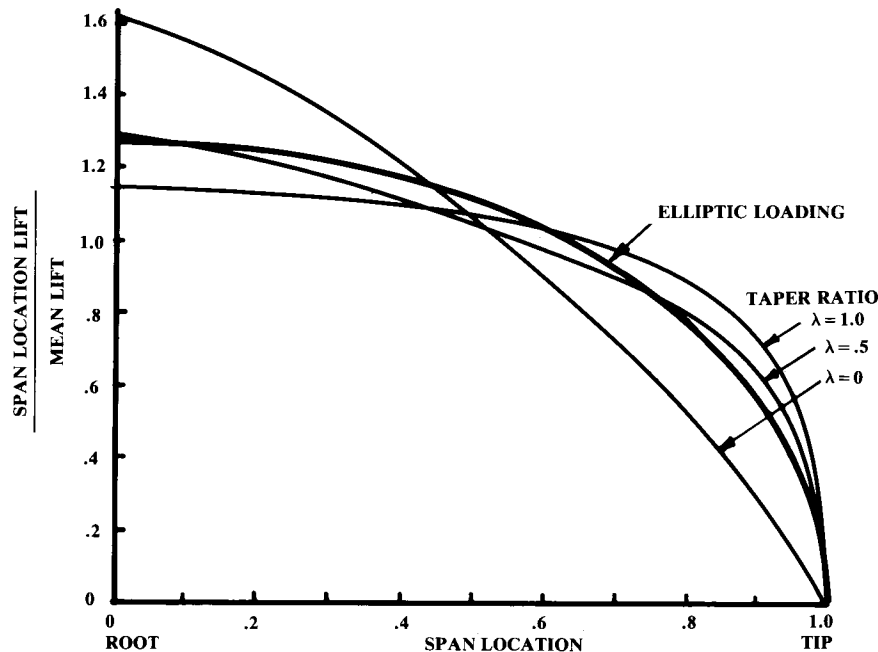


Fig. 4.22 Effect of taper on lift distribution.

form is shaped as an ellipse, as shown in Fig. 4.21. This result was the basis of the graceful wing of the Supermarine Spitfire, a leading British fighter of World War II.

An elliptical wing planform is difficult and expensive to build. The easiest wing to build is the untapered ($\lambda = 1.0$) rectangular wing. However, the untapered wing has constant chord length along the span, and so has excessive chord towards the tip when compared to the ideal elliptical wing. This "loads up" the tip, causing the wing to generate more of its lift toward the tip than is ideal. The end result is that an untwisted rectangular wing has about 7% more drag due to lift than an elliptical wing of the same aspect ratio.

When a rectangular wing is tapered, the tip chords become shorter, alleviating the undesired effects of the constant-chord rectangular wing. In fact, a taper ratio of 0.45 almost completely eliminates those effects for an unswept wing, and produces a lift distribution very close to the elliptical ideal (Fig. 4.22). This results in a drag due to lift less than 1% higher than the ideal, elliptical wing.

A wing swept aft tends to divert the air outboard, towards the tips. This loads up the tips, creating more lift outboard than for an equivalent unswept wing. To return the lift distribution to the desired elliptical lift distribution, it is necessary to increase the amount of taper (i.e., reduce the taper ratio, λ).

Figure 4.23 illustrates the results of NACA wind tunnel tests to determine the taper ratio required to approximate the elliptical lift distribution for a swept untwisted wing. This figure can be used for a first approximation of the desired taper ratio for a swept wing. However, it should be noted that taper ratios much lower than 0.2 should be avoided for all but delta wings, as a very low taper ratio tends to promote tip stall.

Figure 4.23 also indicates that an untwisted wing with no taper should have a forward sweep of 22 deg to approximate an elliptical lift distribution. This unusual planform was the basis of the design presented in the first section as Fig. 2.5. The intent was to provide an elliptical lift distribution with an easy-to-construct rectangular wing.

However, cost analysis indicated that the total reduction in manufacturing cost was small. Furthermore, the weight increase caused by the lack of wing thickness at the root when compared to a conventional, tapered wing caused this design to cost more than a regular design. (Well, at least it was an interesting trade study!)

Twist

Wing twist is used to prevent tip stall and to revise the lift distribution to approximate an ellipse. Typically, wings are twisted between zero and five degrees.

"Geometric twist" is the actual change in airfoil angle of incidence, usually measured with respect to the root airfoil. A wing whose tip airfoil

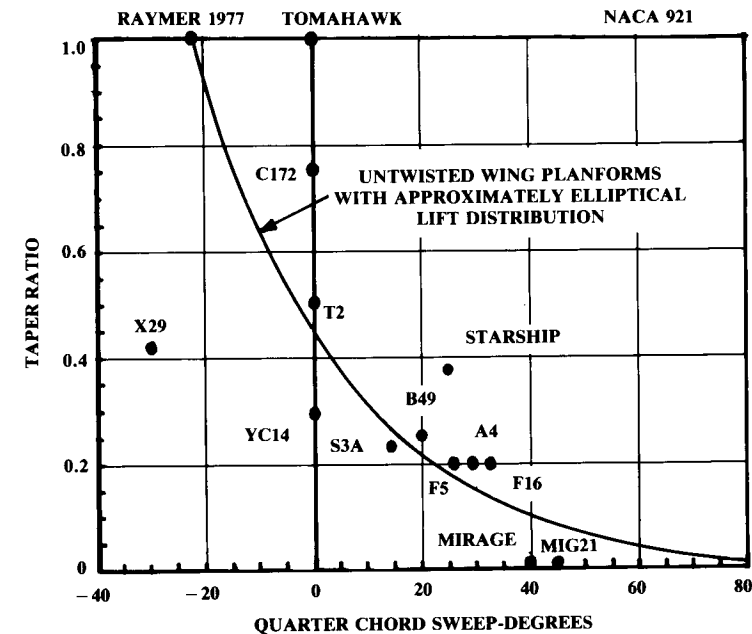


Fig. 4.23 Effect of sweep on desired taper ratio.

is at a negative (nose-down) angle compared to the root airfoil is said to have "wash-out." If a wing has "linear twist," the twist angle changes in proportion to the distance from the root airfoil.

"Aerodynamic twist" is the angle between the zero-lift angle of an airfoil and the zero-lift angle of the root airfoil. If the identical airfoil is used from root to tip, the aerodynamic twist is the same as the geometric twist.

On the other hand, a wing with no geometric twist can have aerodynamic twist if, for example, the root airfoil is symmetric (zero-lift angle is zero) but the tip airfoil is highly cambered (zero-lift angle is nonzero). The total wing aerodynamic twist equals the wing geometric twist plus the root airfoil zero-lift angle, minus the tip airfoil zero-lift angle.

When wing twist is used to reshape the lift distribution, the change in lift at some chord station along the span is proportional to the ratio between the new airfoil angle of attack and the original one. Thus, the effect on lift distribution depends upon the original angle of attack of the wing, which in turn depends upon the lift coefficient at which the wing is flying.

In other words, any attempt to optimize the lift distribution by twisting the wing will be valid only at one lift coefficient. At other lift coefficients, the twisted wing will not get the whole benefit of the twist optimization. The more twist required to produce a good lift distribution at the design lift coefficient, the worse the wing will perform at other lift coefficients. It is for this reason that large amounts of twist (much over 5 deg) should be avoided.

It is very difficult to optimize twist for an arbitrary wing planform. A computerized solution is employed at large companies. For initial design purposes, historical data should be used. Typically, 3 deg of twist provides adequate stall characteristics.

Wing Incidence

The wing incidence angle is the pitch angle of the wing with respect to the fuselage. If the wing is untwisted, the incidence is simply the angle between the fuselage axis and the wing's airfoil chordlines. If the wing is twisted, the incidence is defined with respect to some arbitrarily chosen spanwise location of the wing, usually the root of the exposed wing where it intersects the fuselage. Frequently the incidence is given at the root and tip, which then defines the twist as the difference between the two.

Wing incidence angle is chosen to minimize drag at some operating condition, usually cruise. The incidence angle is chosen such that when the wing is at the correct angle of attack for the selected design condition, the fuselage is at the angle of attack for minimum drag.

For a typical, circular straight fuselage, this is approximately zero degrees angle of attack. For passenger aircraft, the incidence angle must be carefully chosen to insure that the flight attendants do not have to push the food carts uphill!

Wing incidence angle is ultimately set using wind tunnel data. For most initial design work, it can be assumed that general aviation and homebuilt aircraft will have an incidence of about 2 deg, transport aircraft about 1 deg, and military aircraft approximately zero. Later in the design process, aero-

dynamic calculations can be used to check the actual wing incidence angle required during the design condition.

These values are for untwisted wings. If the wing is twisted, the average incidence should equal these values.

A few aircraft have been built with a variable wing incidence angle. The wing aft-attachment is pivoted, and the forward attachment connects to a powerful actuator which pushes the front of the wing up for landing. This arrangement, seen on the Vought F8U Crusader aircraft, allows a short landing gear because the aircraft does not need to rotate to a high fuselage angle for additional lift during takeoff and landing. However, this arrangement is heavy and complicated, and has not been incorporated in a new design in several decades.

Dihedral

Wing dihedral is the angle of the wing with respect to the horizontal when seen from the front. Dihedral tends to roll the aircraft level whenever it is banked. This is frequently, and incorrectly, explained as the result of a greater projected area for the wing that is lowered.

Actually, the rolling moment is caused by a sideslip introduced by the bank angle. The aircraft "slides" toward the lowered wing, which increases its angle of attack. The resulting rolling moment is approximately proportional to the dihedral angle.

Wing sweep also produces a rolling moment due to sideslip, caused by the change in relative sweep of the left and right wings. For an aft-swept wing, the rolling moment produced is negative and proportional to the sine of twice the sweep angle. This creates an effective dihedral that adds to any actual geometric dihedral.

Roughly speaking, 10 deg of sweep provides about 1 deg of effective dihedral. For a forward swept wing, the sweep angle produces a negative dihedral effect, requiring an increased geometric dihedral in order to retain natural directional stability.

In addition, the position of the wing on the fuselage has an influence on the effective dihedral, with the greatest effect provided by a high wing. This is frequently, and incorrectly, explained as a pendulum effect.

Actually, the fuselage in sideslip pushes the air over and under itself. If the wing is high-mounted, the air being pushed over the top of the fuselage pushes up on the forward wing, providing an increased dihedral effect. The reverse is true for a low-mounted wing.

Due to the additive effects of sweep and wing position, many high-winged transports such as the Lockheed C-5 actually require a negative geometric dihedral angle to avoid an excess of effective dihedral. Excessive dihedral effect produces "Dutch roll," a repeated side-to-side motion involving yaw and roll. To counter a Dutch roll tendency, the vertical tail area must be increased, which increases weight and drag.

Unfortunately, as yet no simple technique for selecting dihedral angle takes all of these effects into account. Like so many parameters in initial design, the dihedral angle must be estimated from historical data and then revised following analysis of the design layout.

Table 4.2, developed by the author from data taken from Ref. 1, provides initial estimates of dihedral. For a wing in which the center section is flat and the outer sections alone have dihedral, a first approximation of the required dihedral for the outer panels is the one that places the wing tips as high as they would be for a wing with dihedral starting at the root.

Wing Vertical Location

The wing vertical location with respect to the fuselage is generally set by the real-world environment in which the aircraft will operate. For example, virtually all high-speed commercial transport aircraft are of low-wing design, yet military transport aircraft designed to similar mission profiles and payload weights are all of high-wing design. The reasons for this are discussed later.

The major benefit of a high wing is that it allows placing the fuselage closer to the ground (Fig. 4.24). For military transport aircraft such as the C-5 and C-141, this allows loading and unloading the cargo without special ground handling gear. In fact, these aircraft place the floor of the cargo compartment about 4–5 ft off the ground, which is the height of the cargo area of most trucks. If cargo is needed at a remote field lacking ground-handling gear, the trucks can be backed right up to the aircraft for loading.

With a high wing, jet engines or propellers will have sufficient ground clearance without excessive landing-gear length. Also, the wing tips of a swept high wing are not as likely to strike the ground when in a nose-high, rolled attitude. For these reasons, landing-gear weight is generally reduced for a high-wing aircraft.

Table 4.2 Dihedral guidelines

	Wing position		
	Low	Mid	High
Unswept (civil)	5 to 7	2 to 4	0 to 2
Subsonic swept wing	3 to 7	–2 to 2	–5 to –2
Supersonic swept wing	0 to 5	–5 to 0	–5 to 0

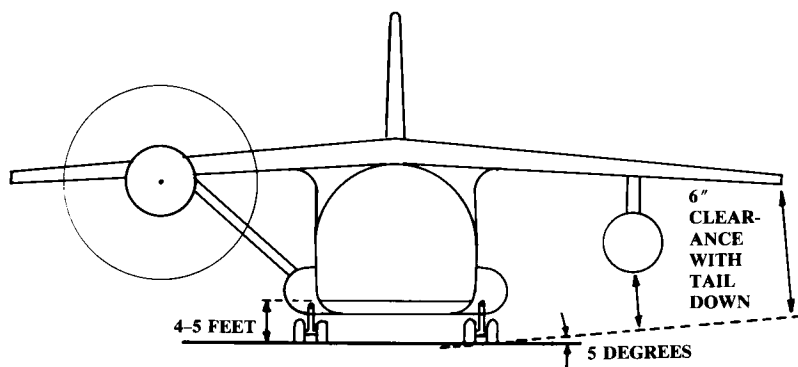


Fig. 4.24 High wing.

For low-speed aircraft, external struts can be used to greatly lower wing weight. However, external struts add substantially to the drag. Since roughly two-thirds of the lift is contributed by the upper surface of the wing, it follows that less drag impact will be seen if the strut disturbs the airflow on the lower surface of the wing than if the strut is above the wing, as would be necessary for a strut-brace, low wing.

Another structural benefit occurs if the wing box is carried over the top of the fuselage rather than passing through it. When the wing box passes through the fuselage, the fuselage must be stiffened around the cut-out area. This adds weight to the fuselage. However, passing the wing box over the fuselage will increase drag due to the increase in frontal area.

For an aircraft designed with short takeoff and landing (STOL) requirements, a high wing offers several advantages. The high position allows room for the very large wing flaps needed for a high lift coefficient. The height of the wing above the ground tends to prevent "floating," where the ground effect increases lift as the aircraft approaches the ground. A floating tendency makes it difficult to touch down on the desired spot. Finally, most STOL designs are also intended to operate from unimproved fields. A high wing places the engines and propellers away from flying rocks and debris.

There are several disadvantages to the high-wing arrangement. While landing-gear weight tends to be lower than other arrangements, the fuselage weight is usually increased because it must be strengthened to support the landing-gear loads. In many cases an external blister is used to house the gear in the retracted position. This adds weight and drag. The fuselage is also usually flattened at the bottom to provide the desired cargo-floor height above ground. This flattened bottom is heavier than the optimal circular fuselage. If the top of the fuselage is circular, as shown in Fig. 4.24, a fairing is required at the wing-fuselage junction.

For small aircraft, the high wing arrangement can block the pilot's visibility in a turn, obscuring the direction toward which the aircraft is turning. Also, the high wing can block upward visibility in a climb. (A classic mid-air collision features a high-wing aircraft climbing into a low-wing one descend-

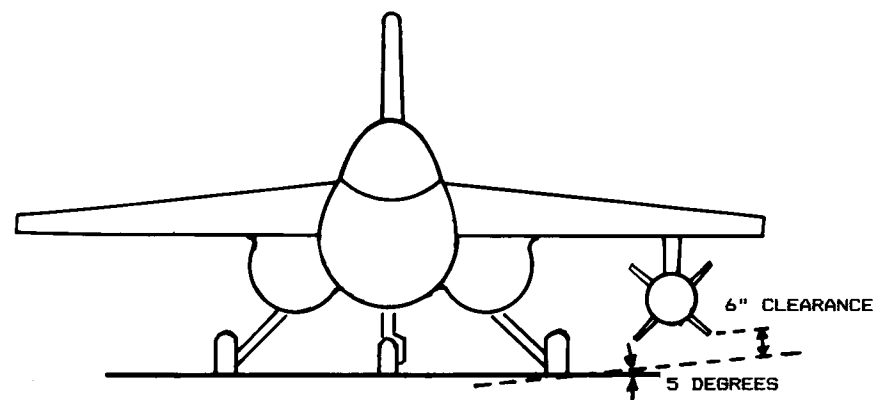


Fig. 4.25 Mid wing.

ing!) Many high-winged light aircraft have transparent panels in the roof to help the pilot see.

If the fuselage is roughly circular and fairings are not used, the mid-wing arrangement (Fig. 4.25) provides the lowest drag. High- and low-wing arrangements must use fairings to attain acceptable interference drag with a circular fuselage.

The mid wing offers some of the ground clearance benefits of the high wing. Many fighter aircraft are mid-winged to allow bombs and missiles to be carried under the wing. A high-wing arrangement would restrict the pilot's visibility to the rear—the key to survival of a fighter in combat.

The mid-wing arrangement is probably superior for aerobatic maneuverability. The dihedral usually required for adequate handling qualities in a low-wing design in normal flight will act in the wrong direction during inverted flight, making smooth aerobatic maneuvers difficult. Also, the effective-dihedral contribution of either high or low wings will make it more difficult to perform high-sideslip maneuvers such as the knife-edge pass.

Structural carrythrough presents the major problem with the mid wing. As will be discussed in Chapter 8, the bending moment produced by the lift on the wing must be carried across the fuselage either by an extension of the wing box ("wing carrythrough box") or by a set of massive ring frames built into the fuselage.

The carrythrough box often proves lighter, but cannot be used in a mid-wing design that must carry cargo or passengers. (One exception to this, the German Hansa executive jet, uses a mild forward sweep to place the carrythrough box behind the passenger compartment.) A carrythrough box is also difficult to incorporate in a mid-wing fighter, in which most of the fuselage will be occupied by the jet engines and inlet ducts.

The major advantage of the low-wing approach (Fig. 4.26) comes in landing-gear stowage. With a low wing, the trunnion about which the gear is retracted can be attached directly to the wing box which, being strong already, will not need much extra strengthening to absorb the gear loads. When retracted, the gear can be stowed in the wing itself, in the wing-fuselage fairing, or in the nacelle. This eliminates the external blister usually used with the high-wing approach.

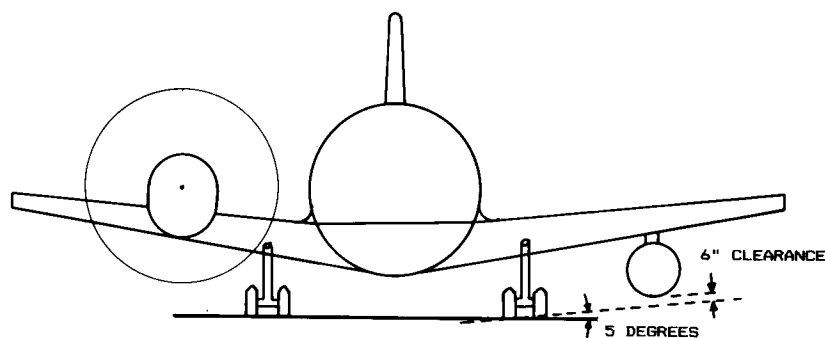


Fig. 4.26 Low wing.

To provide adequate engine and propeller clearance, the fuselage must be placed farther off the ground than for a high-wing aircraft. While this adds to the landing-gear weight, it also provides greater fuselage ground clearance. This reduces the aft-fuselage upsweep needed to attain the required takeoff angle of attack. The lesser aft-fuselage upsweep reduces drag.

While it is true that the low-wing arrangement requires special ground equipment for loading and unloading large airplanes, the high-speed commercial transports are only operated out of established airfields with a full complement of equipment. This is the main reason why military and commercial transports are so different.

Large transports have a fuselage diameter on the order of 20 ft, which allows an uninterrupted passenger compartment above the wing carrythrough box. The wing carrythrough box usually passes through the fuselage for reduced drag, and splits the lower cargo compartment into two compartments. This efficient internal fuselage layout is virtually standard for commercial transports.

If the center wing-panel of a low-wing aircraft lacks dihedral, a one-piece flap which passes under the fuselage can be used. This reduces complexity as well as reducing the risk of asymmetric lift caused by the failure of one flap to extend. Also, the continuous flap will produce more lift and drag than an equal-area flap that is broken at the fuselage.

Several disadvantages of the low-wing approach have already been mentioned, including ground-clearance difficulties. Frequently low-wing aircraft will have dihedral angle set not by aerodynamics, but by the angle required to avoid striking the wing tip on the ground during a bad landing. As was mentioned before, it may require an increase in vertical-tail size to avoid dutch roll with an excessive dihedral angle.

Clearance also affects propellers. To minimize the landing-gear length, many low-wing aircraft have the propellers mounted substantially above the plane of the wing. This will usually increase the interference effects between the wing and propeller, and result in an increase in fuel consumption during cruise.

Wing Tips

Wing-tip shape has two effects upon subsonic aerodynamic performance. The tip shape affects the aircraft wetted area, but only to a small extent. A far more important effect is the influence the tip shape has upon the lateral spacing of the tip vortices. This is largely determined by the ease with which the higher-pressure air on the bottom of the wing can "escape" around the tip to the top of the wing.

A smoothly-rounded tip (when seen nose-on) easily permits the air to flow around the tip. A tip with a sharp edge (when seen nose-on) makes it more difficult, thus reducing the induced drag. Most of the new low-drag wing tips use some form of sharp edge. In fact, even a simple cut-off tip offers less drag than a rounded-off tip, due to the sharp edges where the upper and lower surfaces end. (Fig. 4.27).

The most widely used low-drag wing tip is the Hoerner wingtip (developed by S. Hoerner, Ref. 8). This is a sharp-edged wing tip with the upper surface continuing the upper surface of the wing. The lower surface is

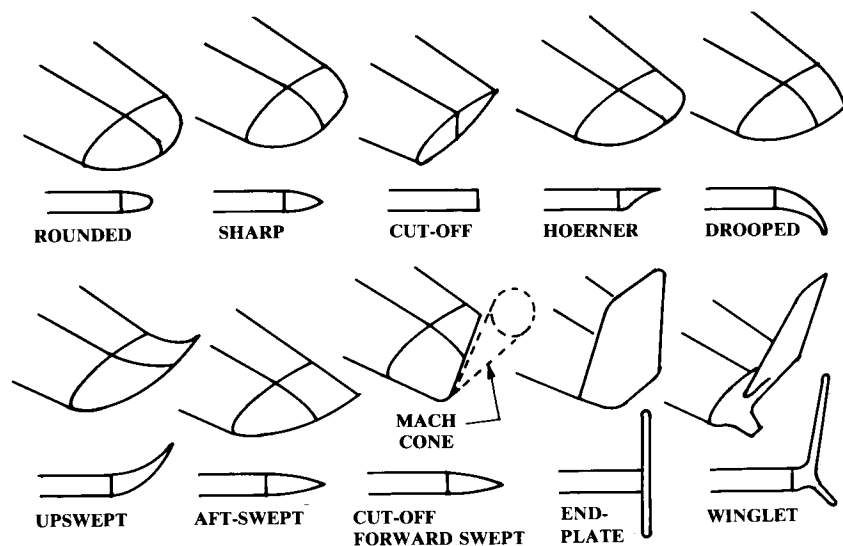


Fig. 4.27 Wing tips.

“undercut” and canted approximately 30 deg to the horizontal. The lower surface may also be “undercambered” (i.e., concave).

The “drooped” and “upswept” wing tips are similar to the Hoerner wing tip except that the tip is curved upwards or downwards to increase the effective span without increasing the actual span. This effect is similar to that employed by endplates, as discussed below.

The sweep of the wing tip also affects the drag. The tip vortex tends to be located approximately at the trailing-edge of the wing tip, so an aft-swept wing tip, with a greater trailing edge span, tends to have lower drag. However, the aft-swept wing tip tends to increase the wing torsional loads.

A cut-off, forward-swept wing tip is sometimes used for supersonic aircraft. The tip is cut off at an angle equal to the supersonic Mach-cone angle, because the area of the wing within the shock cone formed at the wing tip will contribute little to the lift. Also, this tip shape will reduce the torsional loads applied to the wing. The F-15 fighter uses such a cut-off tip for both wings and horizontal tails.

Induced drag is caused by the higher-pressure air at the bottom of the wing escaping around the wing tip to the top of the wing. An obvious way to prevent this would be to mount a vertical plate at the wing tip.

The endplate effect has been known almost since the dawn of flight, but has been seen rarely. The wetted area of the endplate itself creates drag. Also, an endplated wing has an effective span increase of only about 80% of the actual span increase caused by adding the endplates' height to the wing span. However, endplates can be useful when span must be limited.

An advanced version of the endplate can offer lower drag than an equal-area increase in wing span. The “winglet,” designed by NASA's R. Whit-

comb, gets an additional drag reduction by using the energy available in the tip vortex.

The winglet is cambered and twisted so that the rotating vortex flow at the wing tip creates a lift force on the winglet that has a forward component. This forward lift component acts as a “negative” drag, reducing the total wing drag.

A properly designed winglet can potentially provide an effective span increase up to double that bought by adding the winglets' height to the wing span. Winglets provide the greatest benefit when the wing tip vortex is strong, so a low-aspect-ratio wing will see more advantage from the use of winglets than an already-efficient high-aspect-ratio wing.

One problem with winglets is that they add weight behind the elastic axis of the wing, which can aggravate flutter tendencies. Also, the twist and camber of a winglet must be optimized for one velocity. At other than design speed, the winglet will provide less benefit.

For these and other reasons, winglets tend to be used more as add-on devices for existing wings requiring a little more efficiency without major redesign. When an all-new wing is being designed it is usually better to rely upon increased aspect ratio to improve aerodynamic efficiency. This is not always true so a trade study should be conducted sometime during the conceptual design effort.

4.4 BIPLANE WINGS

Biplanes dominated aviation for the first thirty years. The Wright Brothers were influenced by Octave Chanute, a noted architect and civil engineer who applied a structural concept used in bridge building to create lightweight biplane gliders. The early airfoils were thin and birdlike, requiring external bracing, and the biplane arrangement provided more structural efficiency than an externally-braced monoplane.

With the thicker airfoils now in use, the biplane arrangement is mainly reserved for recreational purposes. However, it should be considered whenever low structural weight is more important to the design than aerodynamic efficiency, or when low speed is required without complicated high-lift devices or excessive wing span.

A biplane should theoretically produce exactly half the induced drag of a monoplane with equal span. Induced drag, or drag-due-to-lift, is a function of the square of the lift being generated. If that lift is split evenly between two wings, each wing should have only one-fourth of the drag of the original wing. Therefore, the total induced drag of a biplane should be two-fourths, or one-half of the value obtained with a monoplane of equal span.

Unfortunately, mutual-interference effects prevent the full benefit from being attained. Good design can yield on the order of a 30% reduction in drag-due-to-lift for a biplane when compared to a monoplane of equal span. However, if the total wing area is held constant to provide the same wing loading for biplane and monoplane, and the monoplane has the same wing span as the biplane, then the aspect ratio of the two wings of the biplane must each be double the aspect ratio of the monoplane.

For a typical monoplane aspect ratio of seven, a biplane would need each wing to have an aspect ratio of fourteen to maintain the same total wing area (and wetted area) while attaining the approximately 30% reduction in induced drag claimed above. Also, if the total wing area and span of a biplane and monoplane are identical, the biplane will have chord lengths half as long as the monoplane. Due to the Reynolds number effect upon airfoil drag, an additional penalty will befall the biplane.

If a monoplane were designed with the same total wing area as the biplane but with an aspect ratio the same as each of the wings of the biplane, it would have a span 41% greater (square root of two, minus one) than the biplane. This would provide a net reduction in drag due to lift of about 31% when compared to the biplane of equal wing area ($1 - 0.5/0.7$). Thus, a biplane will actually provide a reduction in induced drag only if the aircraft's total span is limited for some reason to a value less than that desired for a monoplane.

Span can be limited for a number of reasons. For an aerobatic aircraft, a reduced span will increase the roll rate. For an aircraft flying at very low speeds, the wing area required to support the aircraft may require a wing span larger than practical from a structural viewpoint. Span can also be limited by the available hangar width. All of these reasons contributed to the prevalence of the biplane during World War I.

Biplane aerodynamic analysis using Prandtl's interference factor is described in Chapter 12. For initial design purposes, several key concepts should be considered. These are the "gap," "span ratio," "stagger," and "decalage."

Gap is the vertical distance between the two wings. If the gap were infinite, the theoretical result of a halving of the biplane induced drag when compared to an equal-span monoplane would be attained. However, structural weight and the drag of connecting struts generally limit the gap to a value approximately equal to the average chord length. A shorter gap will produce increasing interference between the two wings, raising the overall drag.

Span ratio is the ratio between the shorter wing and the longer wing. If both wings are the same length, the span ratio is one. When span is limited, the minimum induced drag is obtained from equal-length wings. As described, the only technical reason for using the biplane arrangement is the case where span is limited, so the biplane with wings of unequal length should be rarely seen. However, a shorter lower wing has been used in the past to provide better ground clearance.

Stagger is the longitudinal offset of the two wings relative to each other. Positive stagger places the upper wing closer to the nose than the lower wing. Stagger has little or no effect upon drag, and is usually used to improve the visibility upward from a rear-located cockpit. Negative stagger was used in the beautiful Beech D-17 Staggerwing to improve visibility from an enclosed cabin cockpit and to reduce the pitching moment of the large flaps on the lower wing.

Decalage is the relative angle of incidence between the two wings of a biplane. Decalage is positive when the upper wing is set at a larger angle than the lower. In early years much attention was paid to the selection of decalage

to minimize induced drag while encouraging the forward wing to stall before the aft one, thus providing natural stall recovery. Most biplanes since World War I have been designed with zero decalage, although the Pitts Special, holder of numerous world aerobatic championships, has a positive decalage of 1.5 deg.

Much of the discussions above concerning the initial selection of wing geometry can be applied to biplane wings. Most biplanes have wing aspect ratios comparable to monoplanes of similar class (six to eight). As discussed, this yields induced drag levels much higher than obtained from a monoplane with similar wing loading. Taper ratios for biplanes can be selected as for a monoplane, although many biplanes have untapered wings for ease of manufacture.

One or both biplane wings can be swept to enhance stability, improve pilot visibility, or provide room for retractable landing gear. Biplanes typically have dihedral of about 2 deg. Aerobatic biplanes may apply dihedral only to the lower wing.

The mean aerodynamic chord of a biplane can be found as the weighted average of the mean chords of the two wings, weighted by the relative areas of the wings. The biplane aerodynamic center is at approximately 23% of the mean aerodynamic chord, rather than 25% as for a monoplane, due to the wing interference effects.

4.5 TAIL GEOMETRY AND ARRANGEMENT

Tail Functions

Tails are little wings. Much of the previous discussion concerning wings can also be applied to tail surfaces. The major difference between a wing and a tail is that, while the wing is designed routinely to carry a substantial amount of lift, a tail is designed to operate normally at only a fraction of its lift potential. Any time in flight that a tail comes close to its maximum lift potential, and hence its stall angle, something is very wrong!

Tails provide for trim, stability, and control. Trim refers to the generation of a lift force that, by acting through some tail moment arm about the center of gravity, balances some other moment produced by the aircraft.

For the horizontal tail, trim primarily refers to the balancing of the moment created by the wing. An aft horizontal tail typically has a negative incidence angle of about 2–3 deg to balance the wing pitching moment. As the wing pitching moment varies under different flight conditions, the horizontal tail incidence is usually adjustable through a range of about 3 deg up and down.

For the vertical tail, the generation of a trim force is normally not required because the aircraft is usually left-right symmetric and does not create any unbalanced yawing moment. The vertical tail of a multi-engined aircraft must be capable of providing a sufficient trim force in the event of an engine failure.

A single-engine propeller airplane will experience a yawing moment caused by the tail itself. The propeller tends to "drag" the air into a rotational motion in the same direction that the propeller spins. Since the verti-

cal tail is above the fuselage, it will be pushed on by the rotating propwash, causing a nose-left motion for the normal direction of engine rotation. To counter this, some single-engine propeller airplanes have the vertical tail offset several degrees.

The tails are also a key element of stability, acting much like the fins on an arrow. While it is possible to design a stable aircraft without tails, such a design is usually penalized in some other area, such as a compromised airfoil shape, excessive wing area or sweep, or narrow center-of-gravity range.

The other major function of the tail is control. The tail must be sized to provide adequate control power at all critical conditions. These critical conditions for the horizontal tail or canard typically include nosewheel liftoff, low-speed flight with flaps down, and transonic maneuvering. For the vertical tail, critical conditions typically include engine-out flight at low speeds, maximum roll rate, and spin recovery.

Note that control power depends upon the size and type of the movable surface as well as the overall size of the tail itself. For example, several airliners use double-hinged rudders to provide more engine-out control power without increasing the size of the vertical tail beyond what is required for dutch-roll damping. Several fighters, including the YF-12 and the F-107, have used all-moving vertical tails instead of separate rudders to increase control power.

Tail Arrangement

Figure 4.28 illustrates some of the possible variations in aft-tail arrangement. The first shown has become "conventional" for the simple reason

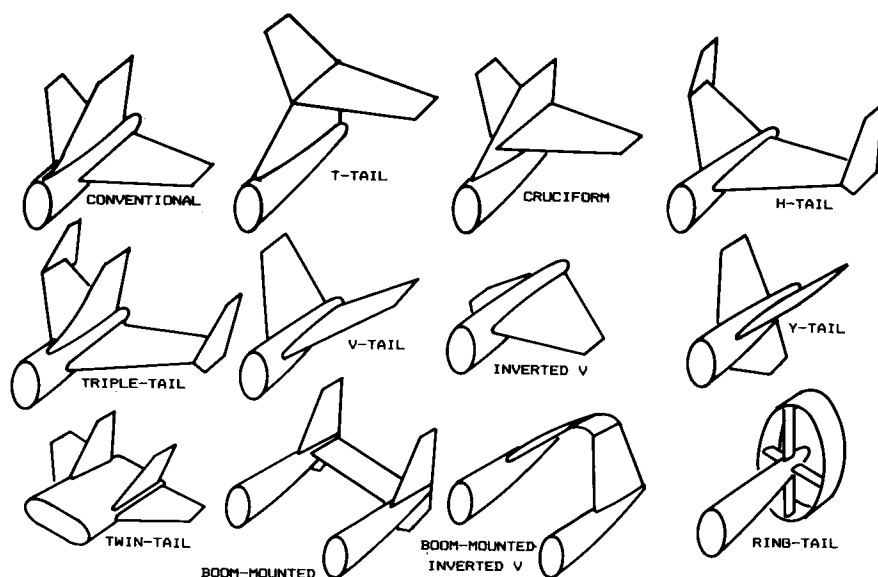


Fig. 4.28 Aft tail variations.

that it works. For most aircraft designs, the conventional tail will usually provide adequate stability and control at the lightest weight. Probably 70% or more of the aircraft in service have such a tail arrangement. However, there are many reasons for considering others.

The "T-tail" is also widely used. A T-tail is inherently heavier than a conventional tail because the vertical tail must be strengthened to support the horizontal tail, but the T-tail provides compensating advantages in many cases.

Due to end-plate effect, the T-tail allows a smaller vertical tail. The T-tail lifts the horizontal tail clear of the wing wake and propwash, which makes it more efficient and hence allows reducing its size. This also reduces buffet on the horizontal tail, which reduces fatigue for both the structure and the pilot.

In jet transport aircraft such as the DC-9 and B-727, the T-tail allows the use of engines mounted in pods on the aft fuselage. Finally, the T-tail is considered stylish, which is not a trivial consideration.

The cruciform tail, a compromise between the conventional and T-tail arrangements, lifts the horizontal tail to avoid proximity to a jet exhaust (as on the B-1B), or to expose the lower part of the rudder to undisturbed air during high angle-of-attack conditions and spins. These goals can be accomplished with a T-tail, but the cruciform tail will impose less of a weight penalty. However, the cruciform tail will not provide a tail-area reduction due to endplate effect as will a T-tail.

The "H-tail" is used primarily to position the vertical tails in undisturbed air during high angle-of-attack conditions (as on the T-46) or to position the rudders in the propwash on a multiengine aircraft to enhance engine-out control. The H-tail is heavier than the conventional tail, but its endplate effect allows a smaller horizontal tail.

On the A-10, the H-tail serves to hide the hot engine nozzles from heat-seeking missiles when viewed from an angle off the rear of the aircraft. H-tails and the related triple-tails have also been used to lower the tail height to allow an aircraft such as the Lockheed Constellation to fit into existing hangars.

The "V-tail" is intended to reduce wetted area. With a V-tail, the horizontal and vertical tail forces are the result of horizontal and vertical projections of the force exerted upon the "V" surfaces. For some required horizontal and vertical tail area, the required V surface area would theoretically be found from the Pythagorean theorem, and the tail dihedral angle would be found as the arctangent of the ratio of required vertical and horizontal areas. The resulting wetted area of the V surfaces would clearly be less than for separate horizontal and vertical surfaces.

However, extensive NACA research (Ref. 3) has concluded that to obtain satisfactory stability and control, the V surfaces must be upsized to about the same total area as would be required for separate horizontal and vertical surfaces. Even without the advantage of reduced wetted area, V-tails offer reduced interference drag but at some penalty in control-actuation complexity, as the rudder and elevator control inputs must be blended in a "mixer" to provide the proper movement of the V-tail "ruddervators."

When the right rudder pedal of a V-tail aircraft is pressed, the right ruddervator deflects downward, and the left ruddervator deflects upward. The combined forces push the tail to the left, so the nose goes to the right as desired. However, the ruddervators also produce a rolling moment toward the left—in opposition to the desired direction of turn, an action called “adverse roll-yaw coupling.”

The inverted V-tail shown in Fig. 4.28 avoids this problem, and instead produces a desirable “proverse roll-yaw coupling.” The inverted V-tail is also said to reduce spiraling tendencies. This tail arrangement can cause difficulties in providing adequate ground clearance.

The “Y-tail” is similar to the V-tail, except that the dihedral angle is reduced and a third surface is mounted vertically beneath the V. This third surface contains the rudder, whereas the V surfaces provide only pitch control. This tail arrangement avoids the complexity of the ruddervators while reducing interference drag when compared to a conventional tail. An inverted Y-tail is used on the F-4, primarily to keep the horizontal surfaces out of the wing wake at high angles of attack.

Twin tails on the fuselage can position the rudders away from the aircraft centerline, which may become blanketed by the wing or forward fuselage at high angles of attack. Also, twin tails have been used simply to reduce the height required with a single tail. Twin tails are usually heavier than an equal-area centerline-mounted single tail, but are often more effective. Twin tails are seen on most large modern fighters such as the F-14, F-15, F-18, and MiG-25.

Boom-mounted tails have been used to allow pusher propellers or to allow location of a heavy jet engine near the center of gravity. Tailbooms are typically heavier than a conventional fuselage construction, but can be desirable in some applications.

Boom-mounted tails can have a mid-mounted horizontal tail or a high horizontal, as on the Cessna Skymaster. Also, the inverted V-tail arrangement can be used with tail booms. The unmanned NASA HiMat research aircraft used boom-mounted verticals with no connecting horizontal tail, instead relying on a canard for pitch control.

The “ring-tail” concept attempts to provide all tail contributions via an airfoil-sectioned ring attached to the aft fuselage, usually doubling as a propeller shroud. While conceptually appealing, the ring-tail has proven inadequate in application. The only recent ring-tail aircraft, the JM-2 raceplane, was ultimately converted to a T-tail.

The location of an aft horizontal tail with respect to the wing is critical to the stall characteristics of the aircraft. If the tail enters the wing wake during the stall, control will be lost and pitchup may be encountered. Several T-tailed aircraft encountered “deep stall” from which they could not be extricated.

Figure 4.29 illustrates the boundaries of the acceptable locations for a horizontal tail to avoid this problem. Note that low tails are best for stall recovery. Also notice that a tail approximately in line with the wing is acceptable for a subsonic aircraft, but may cause problems at supersonic speeds due to the wake of the wing.

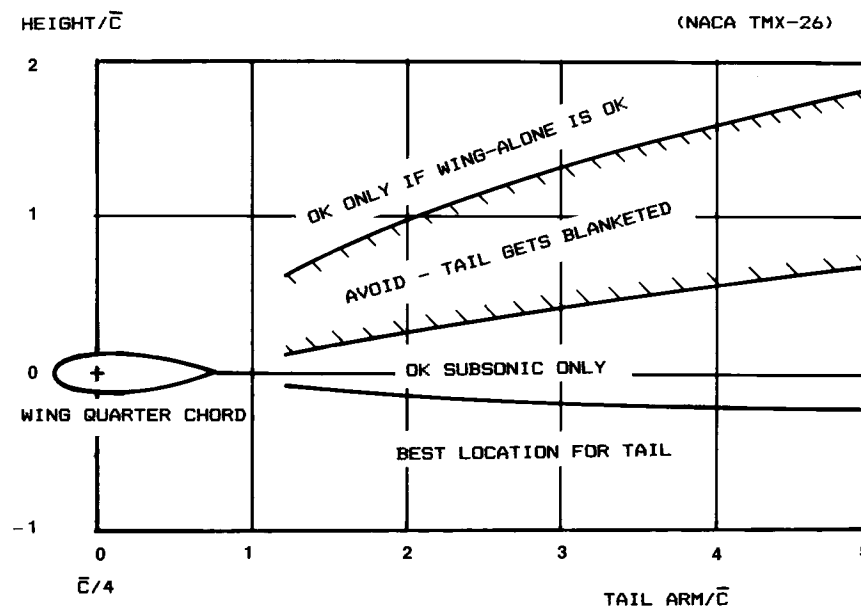


Fig. 4.29 Aft tail positioning.

A T-tail requires a wing designed to avoid pitchup without a horizontal tail, as described by Fig. 4.20. This requires an aircraft stable enough to recover from a stall even when the tail is blanketed by the wing wake. Several general aviation aircraft use this approach, which has the added benefit of a positive warning to the pilot of impending stall caused by buffeting on the tail as it enters the wing wake at high angle of attack.

Other possible tail arrangements are depicted in Fig. 4.30. Canards were used by the Wright brothers as a way of ensuring adequate control power, but fell out of favor due to the difficulty of providing sufficient stability. The early Wright airplanes were, in fact, quite unstable, and required a well-trained pilot with quick reflexes. Movie footage taken by passengers shows the Wright canards being continuously manipulated from almost full-up to full-down as the pilot responded to gusts.

There are actually two distinct classes of canard: the control-canard and the lifting-canard. In the control-canard, the wing carries most of the lift, and the canard is used primarily for control (as is an aft tail). Both the Wright Flyer and the Grumman X-29 are of this type.

The X-29 configuration is highly unstable in pitch with the canard included but is actually about neutral in stability with the canard off. This implies that the canard normally operates at nearly zero angle of attack, and thus carries little of the aircraft's weight. This is accomplished by a sophisticated, computerized flight control system that changes the angle of the canard in response to gusts, much as the pilot of an early Wright aircraft was forced to do manually.

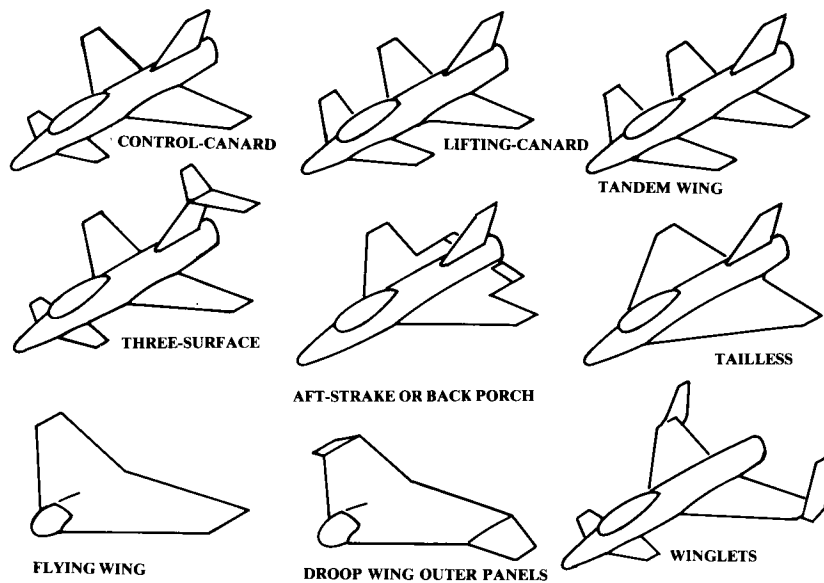


Fig. 4.30 Other tail configurations.

On the X-29 and similar control-canard aircraft, the canard is used to control the angle of attack of the wing and to balance out the pitching moment produced by deflection of the wing flaps.

In contrast, a lifting-canard aircraft uses both the wing and the canard to provide lift under normal flight conditions. This requires that the aircraft center of gravity be well forward of the normal location with respect to the wing when compared to an aft-tailed aircraft. A lifting-canard will usually have a higher aspect ratio and greater airfoil camber than a control-canard, to reduce the canard's drag-due-to-lift.

The lifting-canard arrangement is theoretically more efficient than an aft-tailed aircraft because the canard's lift reduces the lift that must be produced by the wing, which permits a smaller wing and also reduces total drag-due-to-lift. An aft-tail design frequently flies with a download on the tail to produce stability, which actually increases the amount of lift that the wing must produce.

However, the lifting-canard suffers from several drawbacks that reduce the net benefit. First, the use of a lifting-canard has the effect of locating the wing further aft on the aircraft than would be the case with an aft tail. This increases the pitching moment caused by the use of wing flaps.

The pitching moment from the flaps must be balanced during slow flight by a large increase in canard lift. For this reason most aircraft with lifting canards cannot use sophisticated wing flaps. Instead, the wing must be increased in area to provide the same total maximum lift as is provided by the aft-tailed configuration, in which more-powerful flaps may be used. To get around this problem some designs have resorted to the use of slotted canard flaps or even a canard with variable sweep (as on the Beech Starship).

Another drawback is that the canard flap, which is usually used for pitch control, is much closer to the center of gravity than an aft tail elevator would be. This reduces the trim and control effectiveness of the canard flap, requiring a larger surface and producing more trim drag when operating at fairly large deflections such as in slow flight or at an extremely forward center-of-gravity condition. On the other hand, at such conditions an aft tail is producing a download that requires the wing to generate a lift greater than the weight of the aircraft.

One additional benefit obtainable from canards is the avoidance of pitchup. An all-moving canard capable of downward deflections on the order of 45 deg can be used to escape from pitchup situations. This permits optimizing the aspect ratio and sweep without compromising for pitchup avoidance, as suggested by Fig. 4.20.

The tandem-wing is an extension of the lifting-canard concept in which the forward surface produces approximately as much lift as the aft surface. The major benefit of the tandem-wing is a theoretical 50% reduction in the drag-due-to-lift.

The drag-due-to-lift, or induced drag, is a function of the square of the lift being produced. If the weight of the aircraft is evenly distributed to two wings, each wing would have only one-fourth of the induced drag of a single wing. Thus, the sum of the induced drags of the two wings should be half of the drag of a single wing.

As was shown for biplane wings, this theoretical result is not seen in practice because of the interference between the two wings. The second wing must fly in the downwash of the first wing, which requires a higher angle of attack on the second wing. Also, the wake of the first wing tends to create turbulence on the second wing. Finally, to attain stability with a tandem-wing it is usually necessary to move the center of gravity somewhat forward of the location for an even weight split, which may prevent the aft wing from attaining its full lift capability.

To maximize efficiency of a tandem-wing design it is desirable to separate the two wings as far apart as possible, both horizontally and vertically. To attain maximum total lift, it is common to use high-lift devices on the front wing.

A three-surface arrangement provides both aft-tail and lifting-canard surfaces. This allows the use of the lifting-canard for reduction of wing drag-due-to-lift without the difficulty of incorporating wing flaps as seen on a canard-only configuration.

The three-surface aircraft theoretically offers minimum trim drag. A canard or aft-tail, when generating lift for trim purposes, will change the aircraft total lift distribution, which increases total induced drag. On a three-surface configuration the canard and aft-tail can act in opposite directions, thus cancelling out each other's effect upon the total lift distribution. (For example, to generate a nose-up trim the canard can generate an upward lift force while the tail generates an equal downward lift force. The combined effect upon total lift distribution would then be zero.)

However, this reduction in trim drag is a theoretical farfield effect and may not be fully realized in an actual design. The drawback of the three-sur-

face arrangement is the additional weight, complexity, and interference drag associated with the extra surfaces.

The “back-porch” or “aft-strake” is a horizontal control surface that is incorporated into a faired extension of the wing or fuselage. This device, seen on the X-29, is mostly used to prevent pitchup but can also serve as a primary pitch control surface in some cases.

The tailless configuration offers the lowest weight and drag of any tail configuration, if it can be made to work. For a stable aircraft, the wing of a tailless aircraft must be reflexed or twisted to provide natural stability. This reduces the efficiency of the wing.

For an unstable aircraft with a computerized flight control system, this need not be done. In fact, an unstable, tailless aircraft can be designed to be “self-trimming,” meaning that the wing trailing-edge flap angles required to balance the aircraft at different speeds and angles of attack can be designed to be almost exactly the optimal flap angles for maximum L/D .

This is very difficult to accomplish, and is very sensitive to the location of the center of gravity. In fact, all tailless designs are sensitive to center-of-gravity location, and are most successful in designs in which the expendable fuel and payload are located very close to the empty center of gravity.

The vertical tail can also be eliminated for reduced weight and drag. However, the fully tailless (flying-wing) design is probably the most difficult configuration to stabilize, either naturally or by computer. Fully-tailless designs must rely exclusively upon wing control surfaces for control, unless vectored thrust is provided. Rudder control is usually provided by wingtip-mounted drag devices.

Some fully-tailless designs utilize drooped outer wing panels for stability and control enhancement. These act somewhat like an inverted V-tail and provide the desirable proverse roll-yaw coupling with rudder deflection.

Winglets or endplates mounted at the wing tips can be used in place of a vertical tail. This may provide the required vertical tail surface for free, since the effective increase in wing aspect ratio may more than compensate for the wetted area of the tail. To place these tip surfaces far enough aft to act like vertical tails requires either extreme wing sweep or a canard arrangement, or both.

Tail Arrangement for Spin Recovery

The vertical tail plays a key role in spin recovery. An aircraft in a spin is essentially falling vertically and rotating about a vertical axis, with the wing fully stalled. The aircraft is also typically at a large sideslip angle. To recover from the spin requires that the wing be unstalled, so the angle of attack must be reduced. However, first the rotation must be stopped and the sideslip angle reduced, or the aircraft will immediately enter another spin. This requires adequate rudder control even at the high angles of attack seen in the spin.

Figure 4.31 illustrates the effect of tail arrangement upon rudder control at high angles of attack. At high angles of attack the horizontal tail is stalled, producing a turbulent wake extending upward at approximately a 45-deg angle.

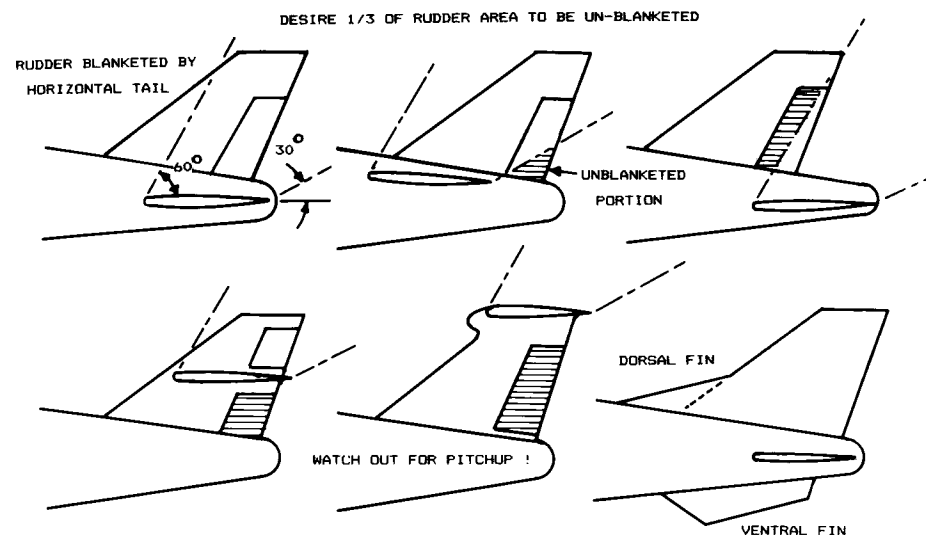


Fig. 4.31 Tail geometry for spin recovery.

In the first example, the rudder lies entirely within the wake of the horizontal tail, so little rudder control is available. The second example shows the effect of moving the horizontal tail forward with respect to the vertical tail. This “uncovers” part of the rudder, improving rudder control. The next example moves the horizontal tail aft with respect to the vertical tail, with the same result. As a rule of thumb, at least a third of the rudder should be out of the wake.

The next two examples show the effect of moving the horizontal tail upward. The T-tail arrangement completely uncovers the rudder, but can result in pitchup.

The last illustration in Fig. 4.31 shows the use of dorsal and ventral fins. The dorsal fin improves tail effectiveness at high angles of sideslip by creating a vortex that attaches to the vertical tail. This tends to prevent the high angles of sideslip seen in spins, and augments rudder control in the spin. The ventral tail also tends to prevent high sideslip, and has the extra advantage of being where it cannot be blanketed by the wing wake. Ventral tails are also used to avoid lateral instability in high-speed flight.

Tail Geometry

The surface areas required for all types of tails are directly proportional to the aircraft’s wing area, so the tail areas cannot be selected until the initial estimate of aircraft takeoff gross weight has been made. The initial estimation of tail area is made using the “tail volume coefficient” method, which will be discussed in Chapter 6.

Other geometric parameters for the tails can be selected at this time. Tail aspect ratio and taper ratio show little variation over a wide range of aircraft types. Table 4.3 provides guidance for selection of tail aspect ratio and

Table 4.3 Tail aspect ratio and taper ratio

	Horizontal tail		Vertical tail	
	A	λ	A	λ
Fighter	3-4	0.2-0.4	0.6-1.4	0.2-0.4
Sail plane	6-10	0.3-0.5	1.5-2.0	0.4-0.6
Others	3-5	0.3-0.6	1.3-2.0	0.3-0.6
T-Tail	—	—	0.7-1.2	0.6-1.0

taper ratio. Note that T-tail aircraft have lower vertical-tail aspect ratios to reduce the weight impact of the horizontal tail's location on top of the vertical tail. Also, some general-aviation aircraft use untapered horizontal tails ($\lambda = 1.0$) to reduce manufacturing costs.

Leading-edge sweep of the horizontal tail is usually set to about 5 deg more than the wing sweep. This tends to make the tail stall after the wing, and also provides the tail with a higher Critical Mach Number than the wing, which avoids loss of elevator effectiveness due to shock formation. For low-speed aircraft, the horizontal tail sweep is frequently set to provide a straight hinge line for the elevator, which usually has the left and right sides connected to reduce flutter tendencies.

Vertical-tail sweep varies between about 35 and 55 deg. For a low-speed aircraft, there is little reason for vertical-tail sweep beyond about 20 deg other than aesthetics. For a high-speed aircraft, vertical-tail sweep is used primarily to insure that the tail's Critical Mach Number is higher than the wing's.

The exact planform of the tail surfaces is actually not very critical in the early stages of the design process. The tail geometries are revised during later analytical and wind-tunnel studies. For conceptual design, it is usually acceptable simply to draw tail surfaces that "look right," based upon prior experience and similar designs.

Tail thickness ratio is usually similar to the wing thickness ratio, as determined by the historical guidelines provided in the wing-geometry section. For a high-speed aircraft, the horizontal tail is frequently about 10% thinner than the wing to ensure that the tail has a higher Critical Mach Number.

Note that a lifting canard or tandem wing should be designed using the guidelines and procedures given for initial wing design, instead of the tail-design guidelines described above.

THRUST-TO-WEIGHT RATIO AND WING LOADING

5.1 INTRODUCTION

The thrust-to-weight ratio (T/W) and the wing loading (W/S) are the two most important parameters affecting aircraft performance. Optimization of these parameters forms a major part of the analytical design activities conducted after an initial design layout. The methods used for this optimization are described in Chapter 19.

However, it is essential that a credible estimate of the wing loading and thrust-to-weight ratio be made before the initial design layout is begun. Otherwise, the optimized aircraft may be so unlike the as-drawn aircraft that the design must be completely redone.

For example, if the wing loading used for the initial layout is very low, implying a large wing, the designer will have no trouble finding room for the landing gear and fuel tanks. If later optimization indicates the need for a much higher wing loading, the resulting smaller wing may no longer hold the landing gear and fuel. While they could be put in the fuselage, this would increase the wetted area and therefore the drag, so the optimization results would probably be no good.

Wing loading and thrust-to-weight ratio are interconnected for a number of performance calculations, such as takeoff distance, which is frequently a critical design driver. A requirement for short takeoff can be met by using a large wing (low W/S) with a relatively small engine (low T/W). While the small engine will cause the aircraft to accelerate slowly, it only needs to reach a moderate speed to lift off the ground.

On the other hand, the same takeoff distance could be met with a small wing (high W/S) provided that a large engine (high T/W) is also used. In this case, the aircraft must reach a high speed to lift off, but the large engine can rapidly accelerate the aircraft to that speed.

Due to this interconnection, it is frequently difficult to use historical data to independently select initial values for wing loading and thrust-to-weight ratio. Instead, the designer must guess at one of the parameters and use that guess to calculate the other parameter from the critical design requirements.

In many cases, the critical requirement for wing loading will be the stall speed during the approach for landing. Approach stall speed is independent of engine size, so the wing loading can be estimated based upon stall speed alone. The estimated wing loading can then be used to calculate the T/W required to attain other performance drivers such as the single-engine rate of climb.